



Gastrointestinal System

7. Abdominal pain	73
8. Gastrointestinal haemorrhage	95
9. Diarrhoea	105
10. Dysphagia	112
11. Nausea and vomiting	117
12. Jaundice	127

Acute abdominal pain

The spectrum of diagnoses manifesting as acute abdominal pain is extensive, ranging from the life-threatening to the innocuous. Effective initial assessment and triage requires the rapid recognition of critically unwell patients requiring urgent intervention by utilizing history, examination and, where appropriate, targeted investigations. Proceed with caution on making diagnoses based purely on history and examination as the specificity in diagnosis based on these alone is often poor. In tandem with targeted investigations, however, the diagnostic yield is far higher, especially for the more serious pathologies.

Chronic/episodic abdominal pain

Chronic abdominal pain is common and challenging to assess. Careful evaluation with targeted investigation is required to exclude organic pathology. Most younger patients will have a functional disorder, such as irritable bowel syndrome (IBS), but this should only be a diagnosis of exclusion. Chronic pelvic pain in women should be carefully investigated for a gynaecological cause before a functional diagnosis is made. In older patients with new, persistent abdominal pain, the priority is to exclude underlying pathology, with a focus on malignancy.

It is standard practice to break down the differential diagnoses of abdominal pain by the nine abdominal regions (Fig. 7.1).

Generalized or umbilical abdominal pain

- Perforation
- Intestinal obstruction
- Mesenteric ischaemia
- Ectopic pregnancy
- Testicular torsion
- Pancreatitis
- Ruptured abdominal aortic aneurysm (AAA)

- Aortic dissection
- Gastroenteritis
- Diabetic ketoacidosis/hypercalcaemia/adrenal crisis (diffuse)
- Constipation
- Nonorganic bowel disease (may affect any region)

Epigastric pain

Abdominal

- Peptic ulcer disease/gastritis/gastroesophageal reflux disease (GORD)
- Gastric or pancreatic cancer
- Pancreatitis
- Aortic aneurysm

Referred pain

- Acute coronary syndrome
- Pneumonia

Right hypochondrium

- Gall bladder disease, e.g., biliary colic, cholecystitis, cholangitis
- Liver disorders, e.g., hepatitis, hepatomegaly, trauma, abscess
- Appendicitis with high appendix
- Subphrenic abscess
- Right lower lobe disease, e.g., pneumonia

Left hypochondrium

- Pancreatitis
- Splenic disease, e.g., rupture, aneurysm
- Subphrenic abscess
- Left lower lobe disease, e.g., pneumonia

Right and left lumbar region including loin pain

- Pyelonephritis/perinephric abscess
- Renal colic, e.g., calculus
- Polycystic kidney disease
- Renal tumour
- Adrenal haemorrhage

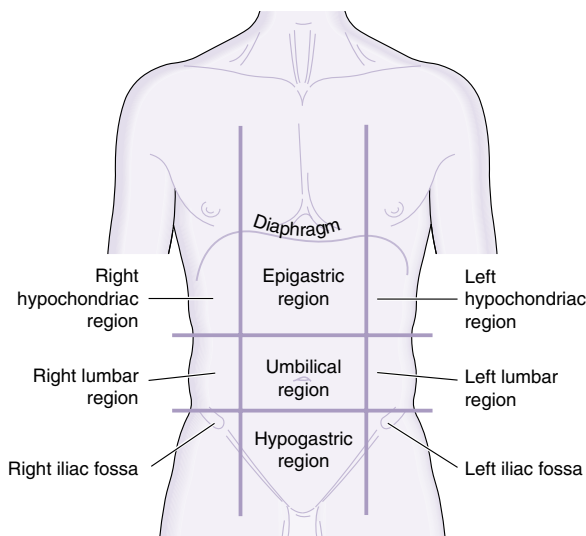


Fig. 7.1 Regions of the abdomen. See text for typical sites of pain. (From Gray J, Smith R, Homer C. Illustrated dictionary of midwifery, 3rd ed. New York: Elsevier; 2022.)

Iliac fossa pain

Right iliac fossa pain

- Appendicitis
- Terminal ileitis, e.g., Crohn's disease, *Yersinia*
- Mesenteric adenitis

Iliac fossa pain on either side

- Diverticulitis
- Colitis
- Colon cancer
- Renal colic
- Ectopic pregnancy or acute ovarian pathology
- Pelvic inflammatory disease/endometriosis
- Ovarian torsion/cyst rupture
- Testicular torsion/trauma/orchiditis
- Hip pathology
- Inguinal hernia
- Psoas abscess

Hypogastric and suprapubic region

- Lower urinary tract infection (UTI)/cystitis
- Urinary retention
- Uterine pathology, e.g., pelvic inflammatory disease, fibroid

Key questions

What are the characteristics of the pain?

Careful consideration of a patient's abdominal pain symptoms will yield immeasurable information to aid targeting investigations. The mnemonic 'SOCRATES' is often helpful: consider the Site, Onset, Character, Radiation, Associated features, Timing, Exacerbating and relieving factors and Severity of the pain.

Visceral pain is conducted by autonomic nerve fibres, so its location corresponds to the embryological origin of the affected structure (Fig. 7.2). The pain may arise from distension or excessive contraction (spasm) of hollow organs. It also arises from tissue damage (inflammation), ischaemia or direct chemical stimulation of pain receptors in organs. It is typically described as dull, is poorly localized and is not associated with abdominal guarding or rigidity. Visceral pain may be constant, e.g., in bowel ischaemia, or 'colicky'. Colicky pain reflects intermittent episodes of intense smooth muscle contraction that produce spasms of discomfort, lasting seconds to minutes before subsiding. Some disorders are described as 'colic' but are actually pseudocolic (e.g., renal or biliary colic). In these cases, the pain builds

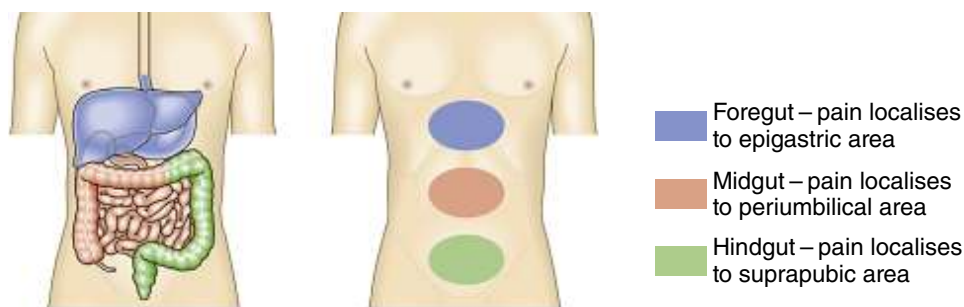


Fig. 7.2 Abdominal pain. Perception of visceral pain is localized to the epigastric, umbilical or suprapubic region, according to the embryological origin of the affected organ. (From Douglas G, Nicol F, Robertson C. *Macleod's clinical examination*, 12th ed. Edinburgh: Churchill Livingstone, 2009.)

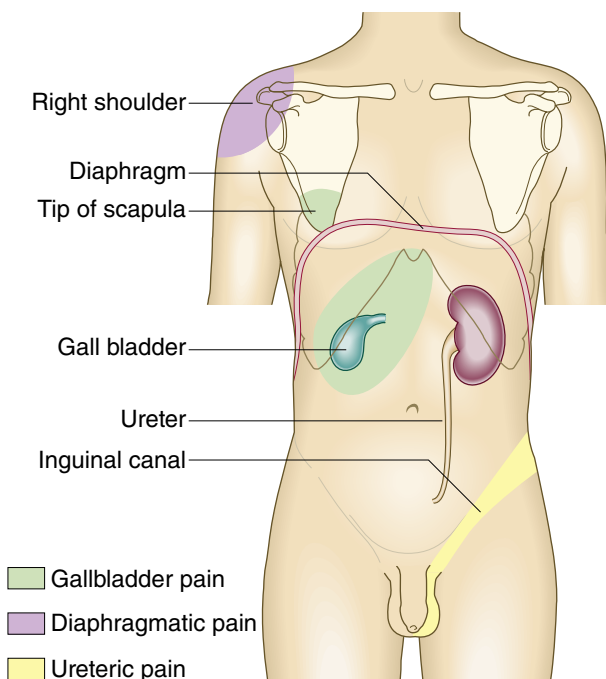


Fig. 7.3 Characteristic radiation of pain from the gallbladder, diaphragm and ureter. (From Douglas G, Nicol F, Robertson C. *Macleod's clinical examination*, 12th ed. Edinburgh: Churchill Livingstone, 2009.)

to a crescendo over several minutes before reaching a steady peak that may last for several hours before easing.

Somatic pain arises from irritation and inflammation of the parietal peritoneum and is conducted by somatic nerves. It is described as sharp, well localized, constant and often associated with local tenderness and guarding. Widespread inflammation of the parietal peritoneum produces generalized peritonitis.

Referred pain is perceived at a site remote from its source and arises due to convergence of nerve fibres at the same spinal cord level (Fig. 7.3).

Is there a systemic inflammatory response?

Many, but not all, serious causes of acute abdominal pain either stem from or provoke an inflammatory process within the abdominal cavity. The presence of fever, $\uparrow$ CRP or

Box 7.1 Indicators of systemic inflammation

- Fever ($>38^{\circ}\text{C}$)
- $\uparrow\text{CRP}$ ($>10\text{ mg/L}$)^a
- $\text{WBC} >11 \times 10^9/\text{L}$ or $<4 \times 10^9/\text{L}$

^aThe significance of the result depends on the clinical context—see text.

$\uparrow\text{WBC}$ with neutrophilia suggests that the patient is mounting an acute systemic inflammatory response (Box 7.1) and may inform the differential diagnosis. In some patients, the presence of inflammatory features may assist the interpretation of uncertain physical signs (e.g., mild localized abdominal tenderness), reinforcing suspicion of local peritonitis. In patients without a clear cause for pain, these features may suggest the need for admission and further investigation. By the same token, the absence of these features can, when used correctly, help to exclude important inflammatory pathology. Finally, recognizing and grading the presence of a systemic inflammatory response are central to the assessment of illness severity.

Note that the significance of an individual result depends on the clinical context, particularly with respect to CRP. In general, the higher the result, the greater the extent of systemic inflammation. A marginal rise in CRP (e.g., $<30\text{ mg/L}$) does not provide compelling evidence of a major inflammatory process. However, if the test is being used to help ‘rule out’ a condition, then it is safer to regard any limit above the upper range of normal as elevated. Caution should be exercised with frail or immunocompromised populations as classic signs of systemic or peritoneal inflammation may be attenuated, resulting in misdiagnosis.

Gastroduodenal disorders

Peptic ulcer disease is a common cause of chronic upper abdominal pain. Almost all duodenal ulcers and 70% of gastric ulcers are attributable to *Helicobacter pylori* infection. The majority of peptic ulcers are asymptomatic. Typical features when they do occur include recurrent episodes of burning or gnawing epigastric discomfort (radiation to

the back may occur); relationship to food (variable); associated dyspeptic symptoms, such as nausea and belching; and relief with antacids or proton pump inhibitors.

Gastritis without frank ulceration may produce similar symptoms.

Gastric cancer occurs more frequently in patients >55 years. In addition to pain, associated symptoms include early satiety, unintentional weight loss and vomiting. All of the above disorders are best diagnosed by upper gastrointestinal endoscopy (UGIE).

Gallstone disease

Gallstones are common in the Western population, and most are asymptomatic.

Biliary colic occurs when the gallbladder contracts, forcing a gallstone to the cystic duct opening, leading to gallbladder distension and spasm. It tends to be postprandial and may be exacerbated by fatty foods, manifesting as intense, dull, RUQ or epigastric pain $\pm$ radiation to the back or scapula (Fig. 7.3). As the biliary tree is not permanently obstructed, it does not cause jaundice, deranged LFTs or abdominal signs. USS should confirm the presence of gallstones or, rarely, demonstrate pathological gallbladder changes. In *cholecystitis*, infection of the gallbladder due to gallstones obstructing the cystic duct occurs. Pain may be accompanied by fever, LFT derangement and RUQ tenderness.

Cholelithiasis (stone in the common bile duct, (CBD)) causes cholestatic jaundice (Chapter 12) with less severe upper abdominal pain, or indeed no pain. In ascending cholangitis, infection of the biliary tree occurs upstream from a blockage in the CBD (gallstone, tumour, liver fluke). Patients present with significant sepsis, jaundice and abdominal discomfort (Charcot triad).

Pancreatic pain

Acute and chronic pancreatitis are often considered as two parts of a disease spectrum of pancreatic disorder. Acute pancreatitis causes severe upper abdominal pain that radiates to the back, often with repeated vomiting. It is associated with a systemic inflammatory response and may progress to multiorgan failure. Diagnosis is made on the

basis of history, examination findings, and blood tests, in particular a raised serum amylase. The majority of cases are caused by gallstones passing down the common bile duct and irritating the pancreas or by alcohol directly injuring the pancreas. Chronic pancreatitis develops in a subset of patients with recurrent episodes of acute pancreatitis. In some patients, the pain is constant and unremitting, while in others, episodes are provoked by drinking alcohol or eating. Associated features relate to the disordered structure and function of the pancreas and include weight loss, anorexia and, in advanced disease, diabetes mellitus (endocrine deficiency) and steatorrhoea (exocrine insufficiency). The majority of cases are due to chronic alcohol excess. Diagnosis is usually made by CT, but endoscopic ultrasound with biopsy may be required to rule out malignancy. Unlike acute pancreatitis, serum amylase is usually unhelpful; ↓faecal elastase indicates pancreatic exocrine insufficiency.

Pancreatic cancer is often painless but may cause unrelenting pain in the upper abdomen that radiates to the back, which is usually associated with cachexia ± cholestatic jaundice.

Bowel ischaemia

Ischaemia of the bowel can be classified into ischaemia of the small bowel (known as mesenteric ischaemia) and ischaemia of the large bowel (known as ischaemic colitis). This can be caused by occlusive (arterial embolism or thrombosis) or nonocclusive (hypoperfusion in septic shock) aetiologies, and both conditions can be acute or chronic.

Mesenteric ischaemia

Acute mesenteric ischaemia presents with acute, severe, constant, diffuse abdominal pain with minimal examination findings (poor localization, no peritonism), systemic upset and, usually, lactic acidosis. Diagnosis can be difficult due to the nonspecific nature of the symptoms. Nevertheless, it is a surgical emergency with a high mortality. 50% of cases are caused by mesenteric embolism (patients may have atrial fibrillation), 25% by thrombosis in situ (atherosclerotic disease), 5% by congestive venous infarction due to

mesenteric vein thrombosis, and 20% by nonocclusive causes such as cardiac failure or septic shock. Diagnosis relies on a high index of suspicion. CT mesenteric angiography is the investigation of choice as plain abdominal CT carries a low sensitivity and specificity.

Chronic mesenteric ischaemia, also called mesenteric angina, is a rare cause of chronic abdominal pain occurring in patients with widespread atherosclerotic disease. Dull periumbilical or lower abdominal pain develops approximately 30 minutes after eating and may be associated with bloody diarrhoea. Weight loss is common as a consequence of food avoidance due to pain or poor absorption. An abdominal bruit may be heard on examination and the diagnosis is made by CT mesenteric angiography.

Ischaemic colitis

Ischaemic colitis is the most common form of bowel ischaemia. The splenic flexure and sigmoid colon are watershed areas between the arterial supply from the superior and inferior mesenteric arteries and therefore susceptible to becoming ischaemic in the presence of hypoperfusion. Ischaemic colitis typically occurs in older patients who present with left-sided abdominal pain and bloody diarrhoea. Diagnosis is confirmed by direct visualization at colonoscopy, but CT is often used as the initial diagnostic test.

Inflammatory bowel disease

Colitis due to either Crohn's disease or ulcerative colitis may cause cramping lower abdominal pain, usually in association with bloody diarrhoea. Small bowel inflammation in Crohn's disease is often manifest by persistent cramping periumbilical or RLQ pain ± diarrhoea (nonbloody) and constitutional upset. Toxic dilatation may occur in both conditions, though is more common with ulcerative colitis and is a surgical emergency. Subacute small bowel obstruction due to oedema or fibrosis causing strictures may lead to colicky postprandial abdominal discomfort. Both disorders are also associated with a range of extraintestinal features (Box 9.3).

Colon cancer

Patients with colon cancer may present with colicky lower abdominal pain from partial or complete bowel obstruction. Other important features include a history of weight loss, change in bowel habit (alternating between constipation and diarrhoea), rectal bleeding and iron deficiency anaemia. Tenesmus is a feature of low rectal tumours. Right-sided cancers present insidiously with vague pain and iron deficiency anaemia. This is because the proximal colon is more distensible and contains liquid faeces, so obstructive features are not seen and blood loss is occult. Screening programmes are in place in many countries due to the outcomes associated with late diagnosis of this common tumour. Tissue diagnosis is usually made by colonoscopy, but CT colonography may be used for frail patients who are unfit for colonoscopy.

Nonorganic bowel diseases

These are extremely common, particularly in younger adults. Diagnosis is based on typical clinical features in the absence of apparent organic disease.

Nonulcer dyspepsia causes symptoms that may be indistinguishable from peptic ulcer

disease. UGIE and mucosal biopsies are normal.

Irritable bowel syndrome (IBS) is a collective term for a group of abdominal symptoms for which no organic cause has been found, including disorders affecting motility and enhancing visceral perception. Diagnostic criteria are shown in [Box 9.1](#). Symptoms tend to follow a relapsing and remitting course, and are often exacerbated by psychosocial stress, menstruation and gastroenteritis. It is essential to make a positive diagnosis after ruling out other organic causes of these symptoms, including inflammatory bowel disease, malignancy, coeliac disease and infection.

Renal tract disorders

Infrequent, discrete attacks of pain $\pm$ haematuria suggest colic caused by renal calculi or clot. Renal colic is severe and may radiate anteriorly and/or to the groin. Classically renal colic causes patients to writhe in pain and they are unable to find a comfortable position. This is in contrast to patients with peritonitis, who lie very still. Chronic dull, aching or 'dragging' discomfort may be due to cancer, adult polycystic kidney disease (APKD), loin pain-haematuria syndrome or chronic

obstruction/pyelonephritis. Frank haematuria requires urological investigation, primarily to rule out malignancy.

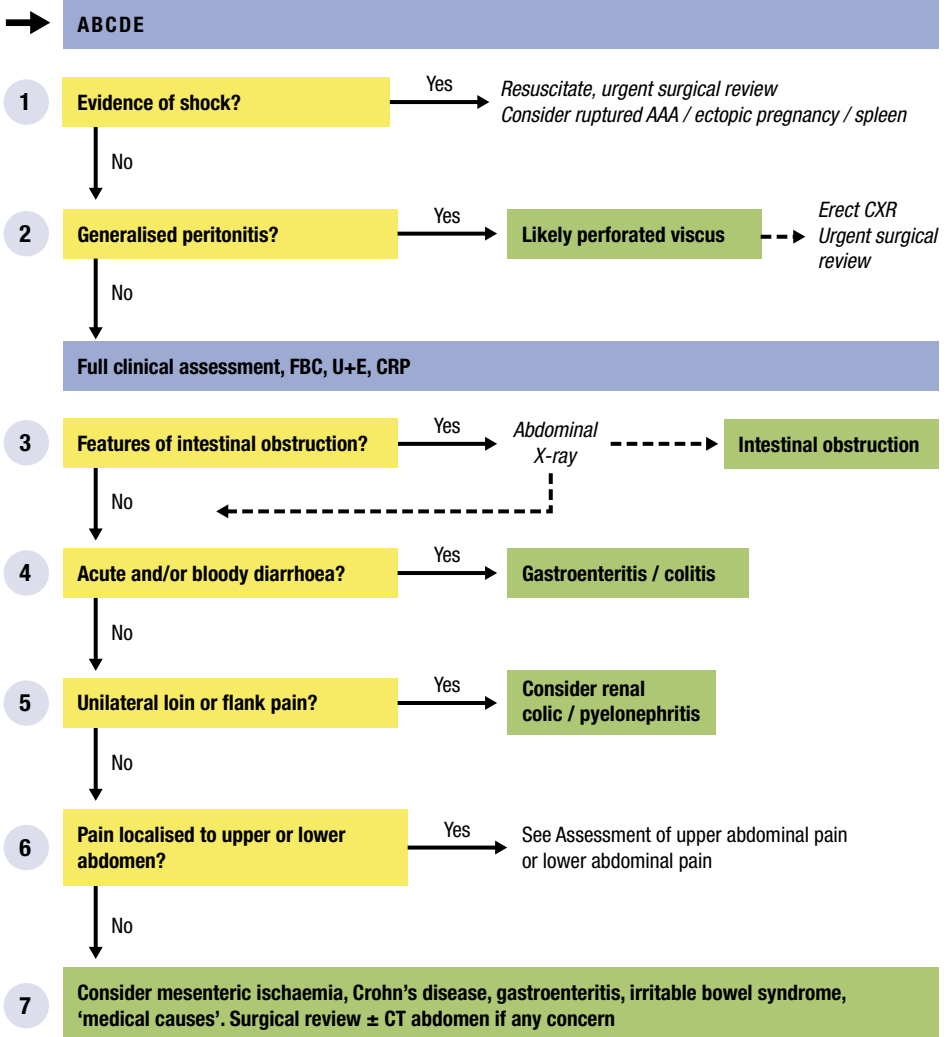
Gynaecological conditions

Abdominal pain in women requires additional consideration, with careful correlation to gynecological history. Sudden-onset lower abdominal pain in women of reproductive age may represent surgical emergencies, such as ovarian torsion or ruptured ectopic pregnancy. Endometriosis (endometrial tissue outside the uterus) and pelvic inflammatory disease (PID) are common causes of chronic lower abdominal pain in women of reproductive

age. Uterine fibroids, ovarian cancer or cysts may present with nonspecific persistent lower abdominal discomfort $\pm$ evidence of a pelvic mass on abdominal/PV examination.

Other diagnostic possibilities

- Constipation
- Small bowel ulcers/tumours
- Atypical infection: giardiasis, abdominal tuberculosis, parasites
- Coeliac disease
- Medical causes of abdominal pain: hypercalcaemia, acute intermittent porphyria, diabetic ketoacidosis (DKA)



1 Evidence of shock?

Rapidly identify patients who are shocked with evidence of hypovolaemia ($\uparrow$ heart rate, $\uparrow$ respiration rate, $\downarrow$ BP) and tissue hypoperfusion (Box 2.1).

If the patient has clinical features of shock, secure two large-bore IV lines; send blood for cross-match, FBC, U+Es, LFTs, coagulation, amylase and arterial/venous blood gas; and begin resuscitation.

The potentially life-threatening diagnoses to consider first include ruptured organs (e.g., AAA, ectopic pregnancy or ruptured spleen) as these require immediate surgical intervention. Peritonism may not be present with these conditions.

- Suspect ruptured AAA in any patient with known AAA, a pulsatile abdominal mass or risk factors (e.g., male >60 years old, positive family history, cigarette smoker) who experiences sudden-onset, severe abdominal/back or loin pain followed rapidly by haemodynamic compromise.
- Suspect ruptured ectopic pregnancy in any pregnant woman or woman of child-bearing age with recent-onset lower abdominal pain or PV bleeding; perform an immediate urine and/or serum hCG test.
- Consider splenic rupture in any shocked patient with abdominal pain who has a history of recent trauma, e.g., road traffic accident.

If any of these diagnoses is suspected, arrange urgent surgical/gynaecological review and appropriate imaging.

In the absence of these conditions, continue to assess for an underlying cause while arranging an ECG, CXR, urinalysis and ABG and refer for surgical review. Other important diagnoses to consider include:

- Perforated viscus.
- Acute mesenteric ischaemia.
- Acute inflammatory conditions, such as pancreatitis, colitis, cholangitis.
- Any condition associated with repeated vomiting, such as intestinal obstruction, gastroenteritis.
- Medical conditions that may 'mimic' a surgical presentation, such as DKA, MI, adrenal crisis, pneumonia.

2 Generalized peritonitis?

Generalized peritonitis is usually caused by leaking enteric fluid from a perforated abdominal viscus, e.g., stomach, duodenum, gallbladder or colon. Pancreatitis may also present in a similar manner. Suspect peritonitis when there is severe, persistent abdominal pain that is worse on movement, coughing or deep inspiration, and generalized abdominal rigidity. The patient will likely be lying still, taking shallow breaths, and will be in obvious distress or discomfort. Inflammatory features are often present but these may be masked/blunted in elderly or immunocompromised patients.

Patients require urgent resuscitation, broad-spectrum antibiotics and immediate surgical referral. Free air under the diaphragm on an erect CXR may be seen (present in 50–75% of cases, Fig. 7.4) though many centres with ready access now consider CT the gold standard due to diagnostic accuracy. Again, maintain a high index of suspicion in the elderly or immunocompromised patient as signs are often subtle, so reassess frequently.

Localized peritonitis occurs where the parietal peritoneum is irritated, for example by an inflamed appendix or diverticular abscess. This results in focal abdominal rigidity over the area affected and may precede progression to generalized peritonitis.



Fig. 7.4 Free air under the diaphragm.

3 Features of intestinal obstruction?

Suspect intestinal obstruction if abdominal pain is colicky and accompanied by vomiting, absolute constipation (no flatus), (no flatus) with or without abdominal distension. The predominant symptoms will vary, depending on the site of obstruction; in high small-bowel obstruction, vomiting and pain are preeminent, whereas in low colonic lesions, constipation and distension are more pronounced. If any of these features are present, perform an abdominal X-ray (AXR, Fig. 7.5) to confirm the diagnosis and estimate the level of the obstruction.

Examine for an incarcerated hernia in any patient with suspected bowel obstruction. Consider further imaging and rectal examination to confirm an obstructing lesion and differentiate it from pseudo-obstruction in patients with large bowel obstruction.

'Drip and suck' is the traditional immediate management: large-bore nasogastric (NG) tube placement to decompress the bowel, and IV fluid and electrolyte replacement. Commence careful monitoring for hypovolaemia and refer to Surgery for further assessment and management.

4 Acute and/or bloody diarrhoea?

Suspect colitis (infective, inflammatory or ischaemic) if the patient has bloody diarrhoea with cramping lower abdominal pain $\pm$ tenesmus and features of systemic inflammation (Box 7.1). If clinically unwell, further imaging in the acute setting may be indicated, particularly if toxic colon is suspected, e.g., inflammatory bowel disease (IBD) or pseudomembranous colitis secondary to *Clostridium difficile*. Otherwise, send stool for culture, ensure appropriate hydration and assess as described in Chapter 9. Always consider the possibility of ischaemic colitis if the patient is elderly or has known vascular disease/atrial fibrillation.

5 Unilateral loin or flank pain?

Consider renal tract obstruction (e.g., stones) in severe, colicky loin pain that radiates to the groin. Patients will typically writhe in pain and be unable to find a comfortable position (in contrast to patients with peritonitis, who lie very still). Visible (macroscopic) or dipstick (microscopic) haematuria is present in 90% of cases and nausea/vomiting is common. AAA can mimic this presentation and should

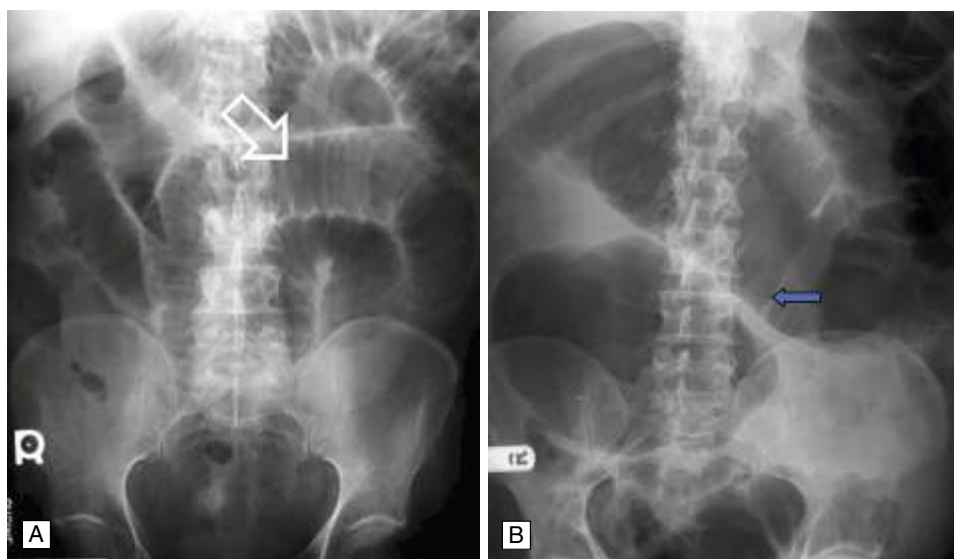


Fig. 7.5 Intestinal obstruction. (A) Small bowel (arrow). (B) Large bowel (arrow). (From Garden OJ, Parks RW, Wigmore SJ. Principles and practice of surgery, 8th ed. New York, Elsevier; 2022.)

be ruled out if a patient is high risk (e.g., male >60 years, positive family history, cigarette smoker) or if the presentation is atypical (e.g., absence of haematuria/restlessness/radiation to groin): arrange an urgent USS and, if this confirms the presence of an AAA, arrange immediate surgical review. Otherwise, organize noncontrast abdominal CT to confirm the presence of a stone (visible in 99% of cases).

In patients with a confirmed stone, check renal function and electrolytes, including acid-base balance, and look for features of infection proximal to the obstruction including ↑temperature/WBC/CRP or leukocytes/nitrites on urinalysis. If you suspect proximal infection, take urine and blood cultures, give IV antibiotics and refer urgently to Urology.

Suspect pyelonephritis if flank pain is non-colicky and associated with inflammatory features (Box 7.1), leukocytes and nitrites (produced by bacteria) on urine dipstick, or loin/renal angle tenderness ± lower urinary tract symptoms. Consider renal USS to exclude a perinephric collection or renal obstruction.

Consider alternative diagnoses (e.g., acute cholecystitis, appendicitis) if there are prominent clinical abdominal signs or if urinalysis is negative for both leukocytes and nitrites. Take blood and urine cultures, start IV antibiotics, arrange US imaging and consult colleagues.

6 Pain localized to upper or lower abdomen?

The localization of pain within the abdomen can be very helpful in narrowing the differential diagnosis (Figs. 7.1 and 7.2).

- If the patient has predominantly RUQ, LUQ, epigastric or generalized upper abdominal pain, proceed to *Acute upper abdominal pain*.
- If the patient has RIF, LIF, suprapubic or bilateral lower abdominal pain, proceed to *Acute lower abdominal pain*.

7 Consider other causes ± surgical review or further imaging if any concern

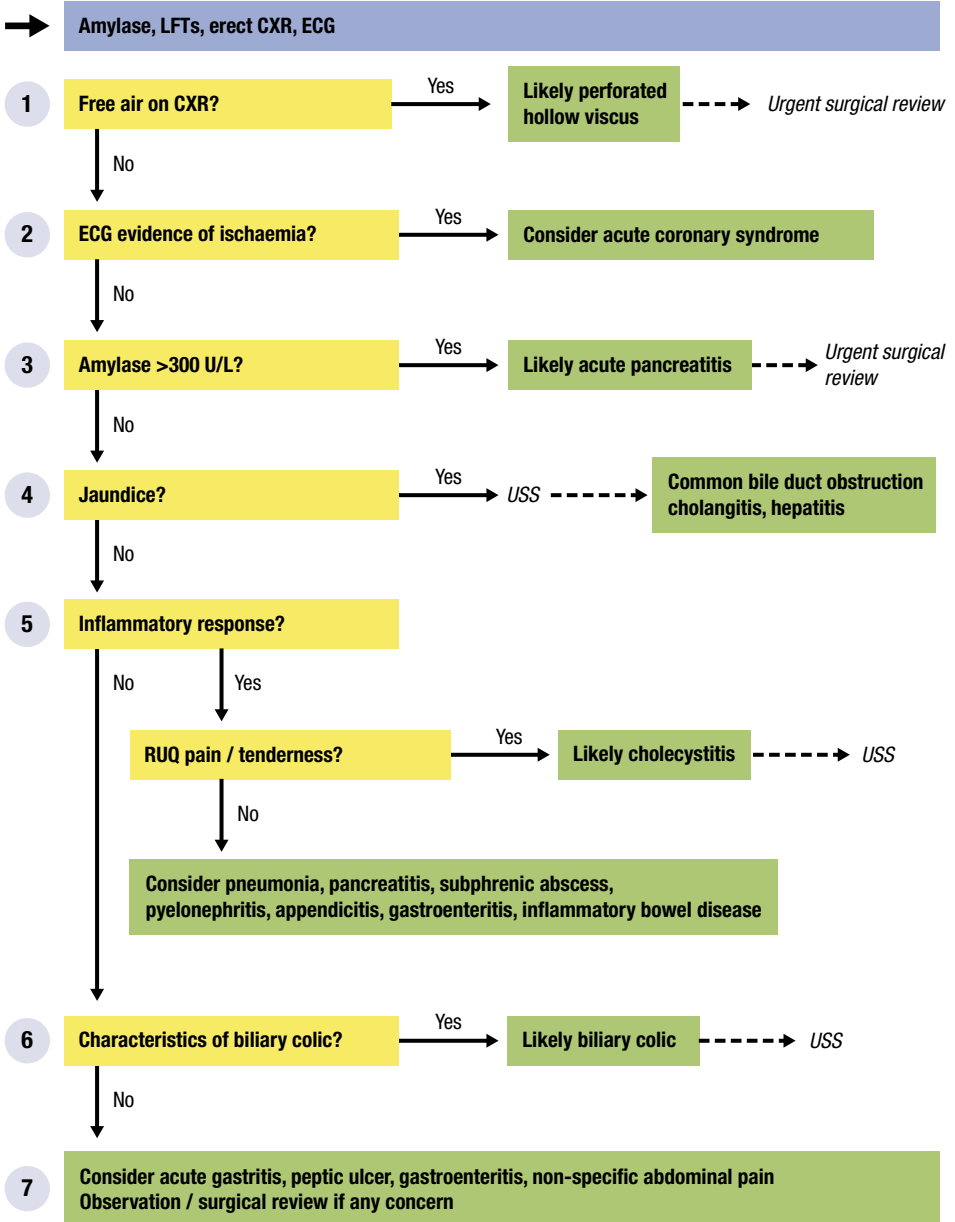
Organize CT angiography to look for features of mesenteric ischaemia in any patient with severe, diffuse pain, shock or unexplained lactic acidosis—especially when patients are elderly or have vascular disease/atrial fibrillation. The abdominal examination may be unremarkable until advanced stages.

Consider atypical presentations of common disorders such as acute appendicitis or inflammatory bowel disease. For example, a retrocaecal appendix may present with flank pain, while any area of the gut may develop a Crohn's inflammatory mass.

Several medical conditions may present with abdominal discomfort, usually associated with conspicuous vomiting and minimal abdominal signs, e.g., DKA, Addison's disease, hypercalcaemia, migraines and MI. It is important to maintain an open mind to these potential causes and undertake appropriate investigations if a surgical cause is not forthcoming.

Nonorganic bowel disease (e.g., IBS or cyclical vomiting) is a frequent cause of acute abdominal pain. The diagnosis of IBS is discussed in Chapter 9, but enquire about a background of longstanding intermittent abdominal pain with altered bowel habit, and review the notes for previous similar admissions.

A period of observation with repeated clinical evaluation is very often the key to successful diagnosis. For example, abdominal pain that was originally central and non-specific may, on repeat examination, have migrated to the RIF, suggesting a diagnosis of acute appendicitis. Patients who remain systemically well and whose pain appears to be settling can usually be discharged safely, with outpatient review. Those with marked systemic upset or other features causing concern but no clear underlying cause require further investigation ± surgical review.



Acute upper abdominal pain: overview

1 Free air on CXR?

Perform an erect CXR in all unwell patients with acute upper abdominal pain; the presence of free air under the diaphragm (Fig. 7.4) indicates perforation of a hollow viscus (present in 50–75% of perforations). Abdominal CT may be the preferred initial diagnostic modality in unwell patients, as long as speed of access doesn't cause delay. Secure wide-bore IV access, cross-match for blood, resuscitate with IV fluids and refer immediately to Surgery.

If a CXR fails to demonstrate free air or is equivocal but clinical suspicion is high (e.g., sudden-onset severe pain with epigastric tenderness and guarding), consider an abdominal CT, but first check amylase and ECG, as described in steps 2 and 3.

2 ECG evidence of ischaemia?

Acute coronary syndromes, particularly inferior myocardial infarction (MI), may present atypically with epigastric pain. Careful clinical consideration is required with this atypical presentation; myocardial ischaemia in patients with stable coronary artery disease may be provoked by severe bleeding, and administration of powerful antithrombotic agents may have catastrophic consequences.

- Perform an ECG in all patients.
- Refer immediately to cardiology if there are features of an ST-elevation MI (Box 3.2).
- In patients with ST depression or other dynamic changes, evaluate carefully for hypotension, sepsis, hypoxia and bleeding before attributing changes to an acute coronary event.
- In patients with nonspecific T-wave changes (Box 3.2), measure cardiac biomarkers to assist diagnosis and continue to search for alternative causes.
- Seek Cardiology input if there is any diagnostic doubt.

3 Amylase >300 U/L?

Measure serum amylase in any patient with acute severe epigastric pain. Patients with an amylase >3× the reference range have a 95% likelihood of having pancreatitis; levels >1,000 U/L are considered diagnostic. Amylase levels are not considered to correlate to disease severity.

If amylase levels are normal or equivocal, continue to suspect the diagnosis if the history is characteristic and:

- There has been a delay in presentation *OR*
- Pancreatic insufficiency is suspected, that is, there is a history of alcoholism, previous episodes of pancreatitis or known chronic pancreatitis.

In these patients, consider CT with IV contrast (provided renal function is satisfactory) to look for evidence of pancreatic inflammation.

This unpredictable disease is associated with a high rate of mortality (12%) and complications. Once the diagnosis has been made, evaluate repeatedly for evidence of complications such as shock, sepsis, hypoxia (ARDS), and disseminated intravascular coagulation (DIC). Calculate the Glasgow prognostic criteria score (Box 7.2) or another validated

7**Box 7.2 Modified Glasgow criteria for predicting severity of acute pancreatitis**

- PaO_2 <8 kPa (60 mmHg)
- Age >55 years
- Neutrophilia WBC > $15 \times 10^9/\text{L}$
- Calcium (corrected, serum) <2.00 mmol/L (8 mg/dL)
- Renal function Urea >16 mmol/L (45 mg/dL) (after rehydration)
- Enzymes ALT >200 U/L or LDH > 600 U/L
- Albumin <32 g/L
- Sugar Glucose >10 mmol/L (180 mg/dL)

**Severity and prognosis worsen as the number of these factors increases. More than three implies severe disease.*

prognostic score. Refer patients with severe or high-risk acute pancreatitis (shock, organ failure, Glasgow score ≥ 3) to a critical care unit. Refer all patients with pancreatitis to Surgery so that further investigation and management can be determined.

4 Jaundice?

Organize an urgent abdominal USS for all patients with acute upper abdominal pain and jaundice to look for evidence of biliary obstruction or hepatitis.

Assume biliary sepsis, at least initially, if the jaundiced patient is unwell with evidence of an inflammatory response (Box 7.1), sepsis or cholestatic jaundice (Chapter 12); give IV antibiotics and, if the USS confirms a dilated CBD, refer immediately to Surgery for further investigation (MRCP) and biliary decompression (ERCP).

Assess as described in Chapter 12 if there are clinical or USS features of acute hepatitis.

5 Inflammatory response?

At this stage, use the presence or absence of a systemic inflammatory response (Box 7.1) to narrow the differential diagnosis.

Arrange prompt abdominal USS to confirm or exclude acute cholecystitis in any patient with inflammatory features accompanied by any of the following:

- Localized RUQ pain.
- Direct tenderness to palpation in the RUQ.
- Positive Murphy sign (sudden arrest of inspiration while taking a deep breath during palpation of the gallbladder).

In the absence of acute cholecystitis, the USS may reveal an alternative cause for the

presentation, such as pyelonephritis, hepatitis or a subphrenic collection.

If none of these features are present, consider alternative disorders:

- Basal pneumonia if there is clinical or CXR evidence of basal consolidation especially if accompanied by productive cough or dyspnoea.
- Gastroenteritis in patients with an acute vomiting illness and no abdominal guarding or rigidity (reassess regularly).
- Acute pyelonephritis if urinalysis is positive for leucocytes/nitrites.

Otherwise, consider USS/CT to exclude atypical presentations of acute appendicitis/pancreatitis/cholecystitis, Crohn's disease or other acute inflammatory pathology.

6 Characteristics of biliary colic?

Biliary colic is a common cause of acute upper abdominal pain in patients who are otherwise well and do not exhibit evidence of a systemic inflammatory response. Abdominal USS can assist the diagnosis by demonstrating the presence of gallstones. However, asymptomatic gallstones are very common, and so the history is critical to making an accurate diagnosis.

Look for the following suggestive features:

- Nocturnal or postprandial pain.
- Duration ≤ 6 hours followed by complete resolution of symptoms.
- The main site is the epigastrium or RUQ $\pm$ radiation to the back.
- Constant, vague, aching or cramping discomfort.
- History of previous similar episodes.

Arrange abdominal USS only in patients with a suggestive history.

Acute upper abdominal pain: step-by-step assessment**7****Consider other causes ± observation/
surgical review if any concern**

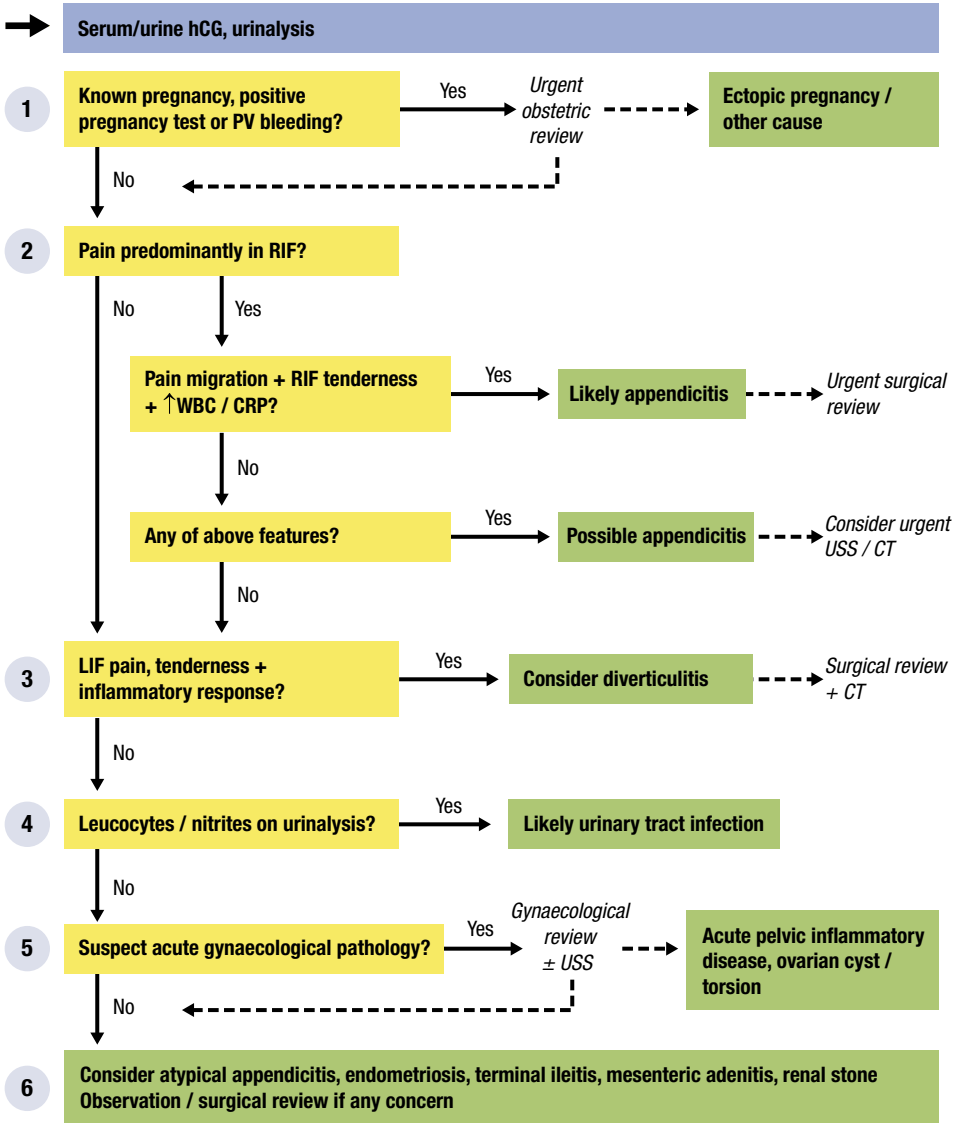
Systemically well patients with an acute vomiting illness and recent infectious contact or ingestion of suspicious foodstuffs are likely to have gastroenteritis.

Suspect acute gastritis if the patient reports new-onset gnawing, burning or vague epigastric discomfort ± mild tenderness, especially if this is associated with dyspeptic symptoms, for example, nausea,

belching, heartburn or a history of recent alcohol excess/NSAID use. Consider peptic ulcer disease if there is a background history of similar symptoms.

In many cases, no definite diagnosis is reached. Admit for observation ± surgical review if symptoms are not improving or there are worrying features on examination. Otherwise, patients can usually be discharged, with further outpatient assessment if symptoms recur or persist.

7



Acute lower abdominal pain: overview

1 Known pregnancy, positive pregnancy test or PV bleeding?

Perform a bedside urine pregnancy test in any pre- or perimenopausal woman with acute lower abdominal pain. If the urine pregnancy test is negative but there is ongoing suspicion of pregnancy (e.g., missed period, vaginal bleeding) send blood for formal laboratory measurement of hCG. If urine or serum hCG are positive, request an urgent gynaecological review with transabdominal ± transvaginal USS to exclude ectopic pregnancy.

Request an obstetric review for assessment of pregnancy-related complications in any woman with known intrauterine pregnancy who develops acute lower abdominal pain, but consider alternative diagnoses, including UTI and acute appendicitis.

2 Pain predominantly in RIF?

Stereotypical clinical characteristics of pain secondary to acute appendicitis are:

- Migration of pain from the periumbilical region to the RIF.
- RIF tenderness or signs of local peritonism.

Several clinical signs may be sought, including Rovsing sign (pain felt in right lower abdomen on palpation of left side) or the psoas sign (pain elicited by lifting the right leg against resistance when in the supine position), but diagnostic accuracy is significantly influenced by a clinician's experience. Systemic upset may also be present (nausea, vomiting, anorexia) alongside evidence of systemic inflammation (Box 7.1). High degree of suspicion should be maintained as clinical signs may be subtle early in the disease course or in frail patients. An ↑WBC and ↑CRP are not specific for appendicitis, but the diagnosis is unlikely if both are within normal limits. Nevertheless, a surgical review is mandatory if the presentation is otherwise

typical—especially if <12 hours from onset of symptoms.

Request prompt surgical review if high index of suspicion and consider urgent imaging to assist diagnosis. Abdominal USS may help to confirm the diagnosis rapidly or, in females, identify pelvic pathology, such as ovarian cyst/torsion. CT offers greater diagnostic accuracy and may be considered in males or if diagnostic uncertainty persists after USS.

3 LIF pain, tenderness + inflammatory response?

Exclude sigmoid diverticulitis in any patient with acute LIF pain and tenderness (± guarding) with evidence of systemic inflammation (Box 7.1)—especially if >40 years. Even in the absence of overt tenderness or inflammatory features, maintain a high degree of suspicion if the patient is elderly or has known diverticular disease. Arrange urgent CT to confirm the diagnosis and identify complications (e.g., abscess) or to seek an alternative cause for the presentation.

4 Leukocytes/nitrites on urinalysis?

Suprapubic pain/tenderness is common in UTI, but the absence of nitrites and leukocytes on urinalysis makes the diagnosis unlikely.

Acute appendicitis may cause dysuria, frequency and urgency with positive urinalysis (e.g., haematuria, leukocytes) if the inflamed appendix lies adjacent to the bladder or ureter; always consider this possibility, particularly in males (in whom cystitis is rare). However, in the absence of other worrying features, a positive urinalysis for both leukocytes and nitrites, especially in women, suggests UTI; send an MSU and start empirical treatment.

5 Suspect acute gynaecological pathology?

Have a high index of suspicion for acute gynaecological pathology in any woman of child-bearing age with acute pelvic or lower abdominal pain. There are no definitive signs for clinical diagnosis. Organize an urgent transabdominal/transvaginal USS in patients <35 years who present with abrupt-onset, severe, unilateral pelvic or lower abdominal pain with associated nausea, adnexal tenderness or a palpable adnexal mass. The principal aim of USS is to look for evidence of ovarian torsion, but it may reveal an alternative diagnosis, such as ovarian cyst. Request a Gynaecology review if you have a strong clinical suspicion of torsion or the USS result is positive/equivocal.

In the absence of features suggesting ovarian torsion, consider acute PID. Suspect the diagnosis if there is bilateral lower abdominal pain and tenderness $\pm$ fever associated with either of the following features:

- Abnormal vaginal or cervical discharge.
- Acute cervical motion, uterine or adnexal tenderness during bimanual vaginal examination.

The diagnosis is often one of exclusion and, in the emergency setting, it is prudent

to seek formal gynaecological input; in difficult cases, diagnostic laparoscopy may be required as imaging studies have low sensitivity. Most laboratory findings are inconclusive in patients with acute pelvic inflammatory disease as many have no systemic inflammatory response, that is, WCC and CRP are within the normal range. A presumptive diagnosis can be made in sexually active females presenting with lower abdominal pain and cervical motion/uterine/adnexal tenderness on examination. Take endocervical swabs for chlamydia and gonorrhoea, and treat in all cases if positive. HIV and syphilis testing should be discussed.

6 Consider other causes $\pm$ observation/surgical review if any concern

Consider atypical presentations of appendicitis resulting from variations in the position of the appendix; e.g., an inflamed appendix that lies within the pelvis may only be tender on rectal examination. Alternative diagnoses in patients with inflammatory features include terminal ileitis and mesenteric adenitis. Seek a formal surgical opinion if the diagnosis is uncertain.

Acute lower abdominal pain: step-by-step assessment

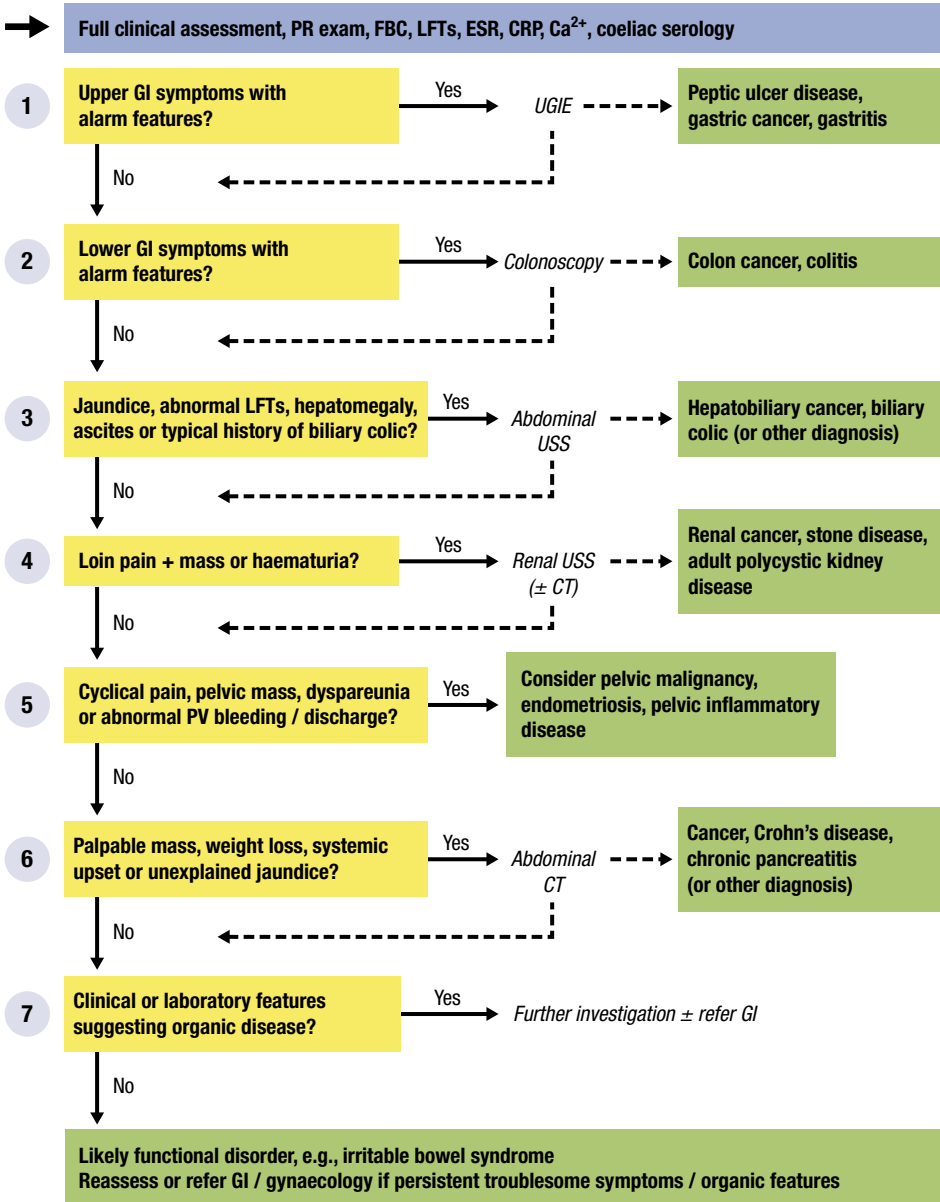
Consider mesenteric ischaemia in any patient who appears unwell or has an unexplained lactic acidosis—especially if they have known vascular disease or atrial fibrillation.

An obstructed renal stone most commonly presents with loin pain but, once descended to the ureter, may cause more localized tenderness in the RIF or LIF; suspect this if the pain radiates to the testes/labia or is associated with visible/dipstick haematuria.

The diagnosis of acute urinary retention is usually obvious but should be excluded in confused patients with lower abdominal tenderness and distress. Clinical examination should be diagnostic but a bladder scan may be helpful.

In female patients, consider the possibility of alternative pelvic pathology, such as endometriosis, especially if there is a relationship to the menstrual cycle.

Chronic/relapsing abdominal pain: overview



Chronic/relapsing abdominal pain: overview

1 Upper GI symptoms with alarm features?

Arrange a prompt UGIE to rule out oesophageal or gastric tumour if there is upper abdominal discomfort with any of the following:

- Weight loss
- Dysphagia
- Persistent vomiting
- Early satiety
- Anorexia
- Haematemesis or melaena
- Iron-deficiency anaemia
- ≥55 years with new-onset persistent symptoms

If a gastric or duodenal ulcer is identified, ensure that a biopsy is negative for malignancy. Prescribe *H pylori* eradication therapy if the CLO test is positive. Assess for risk factors for peptic ulcer disease: recheck the patient's smoking history (smoking is associated with acquisition and increased persistence of *H pylori* infection), alcohol use and medication use (e.g., NSAIDs, bisphosphonates). Confirm successful eradication of *H pylori* with a urea breath test if patient remains symptomatic after 4 weeks.

In patients with macroscopic appearances of an upper GI tumour, obtain the formal biopsy result and discuss urgently with Oncology.

2 Lower GI symptoms with alarming features?

Arrange urgent investigation in any patient with chronic abdominal pain and:

- Rectal bleeding
- Positive faecal occult blood test
- A palpable rectal mass
- Iron-deficiency anaemia
- Change in bowel habit
- Age >45 years
- Recent weight loss

Organize a flexible sigmoidoscopy as the first-line investigation in patients <45 years and a colonoscopy if >45 years. CT colonoscopy may be considered in frail patients or when colonoscopy is unlikely to be well tolerated.

3 Jaundice, abnormal LFTs, hepatomegaly, ascites or typical history of biliary colic?

The combination of new-onset jaundice and persistent/recurrent abdominal discomfort suggests hepatitis, choledocholithiasis or malignancy, e.g., biliary, ampullary, pancreatic or hepatocellular cancer or biliary metastases. Conduct blood testing for liver disorders (Box 7.3) and a transabdominal USS. If the above is inconclusive, further investigation (e.g., CT or MRCP) is mandatory. Discuss with a GI specialist.

Arrange USS and perform a diagnostic ascitic tap in any patient with ascites (send for biochemistry, microbiology and cytology). If the ascites is exudative (serum-ascites albumin gradient (SAAG) <1.1 g/dL) and USS fails to reveal an underlying cause, arrange an abdominal CT scan.

The features of biliary colic are described above. If present, arrange abdominal USS to look for gallstones. Note that gallstones are a very frequent finding in asymptomatic patients so, unless the history is typical or there is evidence of a complication (e.g., dilated CBD, chronic cholecystitis), they are unlikely to explain the presentation.

Refer to a GI specialist if there is a convincing history of biliary pain in the absence of gallstones, especially if associated with abnormal LFTs or a dilated CBD.

4 Loin pain + mass or haematuria?

You must exclude upper renal tract cancer in any patient presenting with persistent loin

Box 7.3 Blood tests to assess liver disorders

- ALT, AST, ALP, GGT
- Albumin
- Viral serology for hepatitis A, B, C
- Ferritin
- Immunoglobulins
- Autoantibodies (ANA/AMA/SMA)
- Alpha-fetoprotein
- Caeruloplasmin
- Alpha-1 antitrypsin

Chronic/relapsing abdominal pain: step-by-step assessment

pain associated with a mass or haematuria (visible or dipstick). Renal USS is a first-line investigation and may reveal alternative causes, such as APKD or chronic hydronephrosis. Consider CT $\pm$ Urology referral if the cause remains unclear.

5 **Cyclical pain, pelvic mass, dyspareunia or abnormal PV bleeding/discharge?**

Arrange a pelvic USS in any patient with a pelvic mass or any postmenopausal patient with PV bleeding or recent onset, persistent lower abdominal pain. The features of PID are described above.

Consider endometriosis in any woman of reproductive age with chronic lower abdominal pain and any of the following features:

- Variation with menstrual cycle
- Deep dyspareunia
- Other prominent cyclical or perimenstrual symptoms

Establishing a definitive diagnosis of endometriosis may be challenging—refer to Gynaecology for further assessment.

6 **Palpable mass, weight loss, systemic upset?**

At this stage, organize an abdominal CT if the patient has weight loss, systemic upset or any palpable masses. Patients with a palpable mass may have been investigated with other modalities in steps 1–5 above, but CT is more likely to identify certain masses (e.g., pancreas, small bowel, renal tract and omental) and should be performed if the cause remains uncertain. In patients with abdominal pain and significant weight loss or other major constitutional upset (e.g., fevers, night

sweats, $\uparrow$ ESR), CT may reveal evidence of lymphoma, solid-organ malignancy or inflammatory disease, for example, chronic pancreatitis, abscess, Crohn's disease.

7 **Clinical or laboratory features suggest organic disease?**

Further investigation is usually required if the patient is >45 years with recent-onset symptoms, weight loss, constitutional upset or abnormal screening investigations. Important organic causes of pain that may easily be missed include chronic pancreatitis, chronic mesenteric ischaemia and Crohn's disease.

Consider chronic pancreatitis in any patient with a background of chronic alcohol excess or steatorrhoea. Check faecal elastase, perform an abdominal CT and consider specialist referral for further assessment.

Exclude mesenteric ischaemia (CT mesenteric angiography) if there is a close relationship between eating and the onset of pain, especially if there is evidence of vascular disease elsewhere, such as angina or intermittent claudication.

Assess for IBD in any patient with chronic abdominal pain and associated diarrhoea. Check faecal calprotectin to assess for luminal inflammation and, if positive, discuss with Gastroenterology about further investigation.

If none of the above is present, a functional cause (e.g., IBS) is likely—particularly when typical symptoms are present ([Box 9.1](#)). It is important to make a positive diagnosis and provide initial management. Patients with persistent, troublesome symptoms may require specialist assessment and investigation to exclude organic disease.

Haematemesis describes vomiting blood and indicates bleeding from the upper GI tract. The appearance of blood in the vomit can vary; fresh, red blood is consistent with a brisk, active bleed; altered blood with a dark, granular appearance ('coffee-grounds') suggests the blood has been partially digested and is consistent with a slower or intermittent bleed. Melaena is the passage of black, tarry stools with a characteristic smell; it is usually due to upper GI bleeding but occasionally due to bleeding within the small bowel or right side of the colon.

Haematochezia is the passage of fresh red or maroon blood per rectum; it is usually due to colonic bleeding, but 15% of cases result from profuse upper GI bleeding.

Haematemesis, melaena or haematochezia in the presence of shock imply active bleeding and are medical emergencies.

Haematemesis

With major upper GI haemorrhage, resuscitation to compensate for ongoing blood loss determines survival and must take place alongside assessment. All hospitals have a major haemorrhage protocol with which you should be familiar. Ensure adequate resuscitation while arranging urgent upper GI endoscopy (UGIE) so that the source of bleeding can be identified and the bleeding stopped.

Peptic ulcer disease

Peptic ulcer disease is the most common cause of upper GI bleeding (estimated 50%) and an important cause of massive haemorrhage. It

is often accompanied by dyspepsia and/or epigastric pain. Important aetiological factors include *Helicobacter pylori* infection, medications (e.g., NSAIDs/aspirin), gastro-oesophageal reflux disease (GORD) and excess alcohol consumption.

Oesophagitis

An increasing proportion of upper GI bleeding is due to oesophagitis. This is usually due to GORD, but other risk factors include medications (e.g., NSAIDs, bisphosphonates, tetracyclines) or infections (e.g., *Candida*, Herpes simplex). There may be a history of heartburn, indigestion or painful swallowing. Upper GI bleeding secondary to oesophagitis is associated with fewer complications and less mortality than bleeding secondary to peptic ulcer disease.

Mallory–Weiss tear

This is a longitudinal mucosal tear of the distal oesophagus and/or proximal stomach following increased gastric pressure. 90% of these bleeds stop spontaneously. The history is characteristic: forceful retching with initially non-bloody vomit, followed by haematemesis.

Portal hypertension with varices

Portal hypertension describes increased pressure within the portal venous system. In Western countries, the most common cause is cirrhosis, most frequently associated with excessive alcohol consumption. However, some non-cirrhotic conditions are associated with portal hypertension, such as schistosomiasis, portal

Differential diagnosis

vein thrombosis or idiopathic portal hypertension. Collateral veins form in response to portal hypertension, allowing portal blood to bypass the liver and enter the systemic circulation directly, for example, at the gastro-oesophageal junction. These oesophageal/gastric varices are superficial and liable to rupture, causing massive GI haemorrhage. Bleeding can be catastrophic, particularly if coagulopathy and thrombocytopenia are present. It is difficult clinically to distinguish variceal bleeding from other causes of massive upper GI bleeding (e.g., peptic ulcer disease), even if the patient has known cirrhosis.

Upper GI malignancy

In patients with upper GI malignancy, there is often a background of weight loss, anorexia, early satiety and/or dysphagia. Rarely, a palpable epigastric mass or signs of metastatic disease may be evident. Risk factors include alcohol, smoking, chewing tobacco, hot drinks, obesity and reflux oesophagitis/Barrett oesophagus.

Other uncommon causes

An aortoduodenal fistula will usually present with massive haematemesis, and this should be suspected if the patient has had previous surgery for an abdominal aortic aneurysm. Angiodysplasia and congenital malformations of the vascular tree (e.g., Dieulafoy lesion) can produce major haemorrhage.

Rectal bleeding

Rectal bleeding is most commonly due to a benign cause. In acute rectal bleeding, similar to upper GI bleeding, assess bleeding severity and ensure adequate resuscitation while arranging interventions to identify and treat the bleeding source. The pattern of bleeding can inform the clinician to the likely location. Anorectal bleeding typically presents with intermittent episodes of minor, fresh, bright red bleeding during or after defecation, which is not mixed with the stool. Distal colon bleeding is darker red and may be partially mixed with stool; proximal colon

bleeding is dark red and fully mixed with stool. Remember that large volume upper GI or proximal colon bleeding can present with fresh bright red bleeding (15% of patients presenting with severe PR bleeding will have an upper GI source).

Diverticular disease

Diviculosis is the most common cause of severe acute rectal bleeding (15–42% of presentations). It typically occurs in older patients. Bleeding occurs due to erosion of a vessel in the neck of a diverticulum and is mostly fresh blood. It may result in a life-threatening bleed but stops spontaneously in 75% of patients. The long-term rebleeding rate is, however, high; up to 40% of patients who undergo conservative management of their initial bleed will rebleed at some point.

Perianal disorders

These are a common cause of rectal bleeding in all age groups. Bright red bleeding is often associated with other anorectal symptoms, such as discomfort, mucus discharge or pruritus. Causes include haemorrhoids, anal fissures, perianal Crohn's disease and anal cancer. Be aware that haemorrhoids can cause major bleeds. Always examine the anus of a patient presenting with rectal bleeding: perianal disorders are diagnosed by PR examination ± proctoscopy. Severe pain, especially during defecation, suggests anal fissure, and the patient will not tolerate PR examination. Always consider a concurrent proximal bleeding source.

Colorectal carcinoma

Colorectal carcinoma is a common cancer. The presentation is frequently insidious with minimal symptoms; be suspicious of recent weight loss, tenesmus, a change in or alternating bowel habit or colicky lower abdominal pain. Have a high degree of suspicion in older adults or those with a genetic predisposition/family history. Bleeding associated with colorectal cancer is usually low volume but recurrent. Polyps may cause rectal bleeding.

Diagnosis is made by colonic imaging (sigmoidoscopy/colonoscopy or CT colonography in patients who are unfit for endoscopic investigation.)

Inflammatory bowel disease

Colorectal inflammation may cause rectal bleeding, with this symptom being more common in ulcerative colitis than in Crohn's disease. The passage of frank blood $\pm$ mucus and tenesmus may occur in proctitis. Colitis produces bloody diarrhoea with intermittent cramping lower abdominal pain and, often, systemic upset. Endoscopy and biopsy of the colon/terminal ileum confirm the diagnosis; inflammatory markers (e.g., CRP, faecal calprotectin) provide a useful guide to disease activity. Chronicity of inflammatory bowel disease can also change the differential for PR bleeding chronic. Perianal Crohn's disease is often associated with complex perianal fistulas.

Ischaemic colitis

Ischaemic colitis (colonic ischaemia) is the most common form of ischaemic injury to the GI tract and can occur because of vascular occlusion (e.g., mesenteric artery embolism or thrombosis) or nonocclusive causes (e.g., hypoperfusion). The superior mesenteric artery supplies the ascending and transverse colon, while the inferior mesenteric artery supplies the descending and sigmoid colon. The splenic flexure and sigmoid colon are areas where the two circulations meet and are considered watershed regions where ischaemic damage is more likely in the presence of hypoperfusion. Clinical presentation varies and depends on the underlying cause. Commonly, ischaemic colitis occurs in older patients, typically resulting in severe lower abdominal pain associated with rectal bleeding. It can affect younger patients, especially

those with known hypercoagulopathy. CT is often used as the initial diagnostic test, but colonoscopy is considered the gold-standard for direct visualization and confirmation of ischaemic colitis. Most cases of ischaemic colitis will improve with treatment of the underlying cause and volume replacement. A small proportion of patients (approximately 20%) will require surgery and resection of the affected bowel.

Angiodysplasia (arteriovenous malformation)

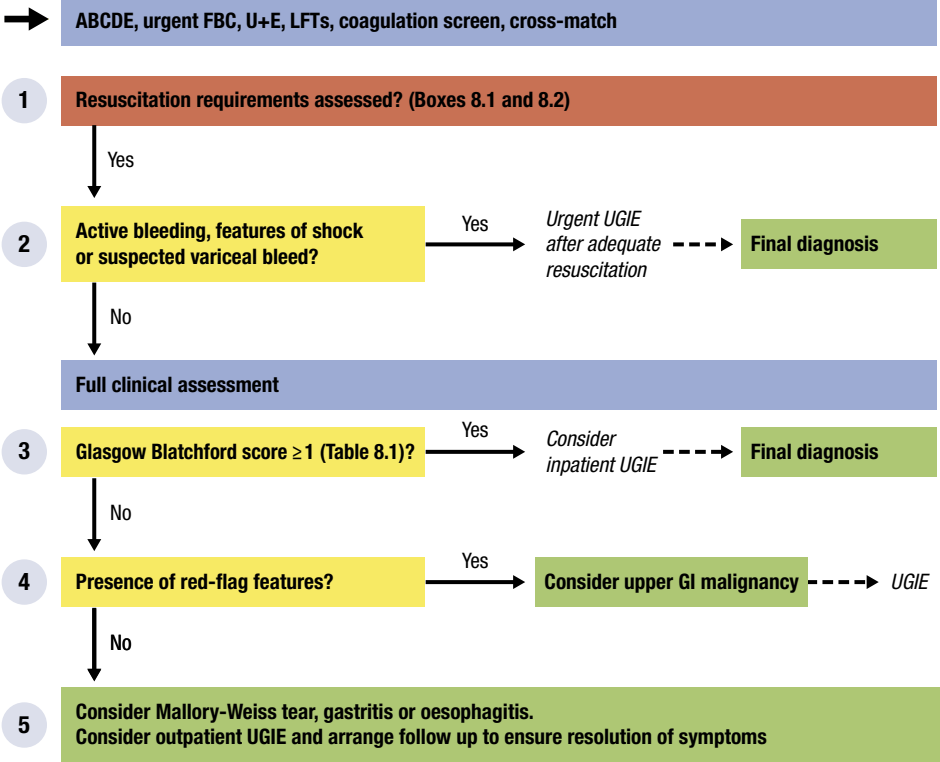
These dilated tortuous submucosal vessels are degenerative vascular malformations and are an important cause of severe lower GI bleeding, predominantly in the elderly. Typical endoscopic features may be apparent, but diagnosis is often challenging. Similar to diverticulosis, bleeding often resolves spontaneously but frequently recurs.

Other causes

GI infections (e.g., *Salmonella*, *Shigella*, *Campylobacter* and *Escherichia coli*) may cause bloody diarrhoea by causing local invasion of the mucosa, usually with systemic upset. Other small intestinal sources of rectal blood loss include Meckel's diverticulum (melaena) and intussusception ('currant jelly' bleeding) in young patients. Acute small bowel ischaemia may be associated with rectal bleeding, but the dominating feature is severe abdominal pain, and the patient is critically unwell. Small bowel tumours are rare and present with intermittent bleeding (may require capsule endoscopy).

Radiation proctocolitis should be considered in any patient with a history of pelvic radiation, such as for prostatic or gynaecological malignancy; most present within 2 years of radiotherapy.

Haematemesis: overview



Haematemesis: overview

Box 8.1 Resuscitation in acute haemorrhage**First steps**

- Recheck airway frequently, especially if ↓GCS/vomiting.
- Give high concentration O₂.
- Get help: call the medical emergency or cardiac arrest team.

Secure IV access and monitoring

- Insert 2 large-bore cannulae in antecubital veins or equivalent.
- If unable to obtain IV access quickly, consider intraosseous access early and get expert help to arrange definitive central venous access.
- Send urgent FBC, U+E, LFTs, coagulation screen, calcium, ABG/VBG and crossmatch, and inform blood bank.
- Monitor vitals and reassess peripheral perfusion every 10–15 min.
- Insert urinary catheter; monitor hourly urine output.

Resuscitate and reassess**Early shock features (Box 8.2) or any ongoing blood loss:**

- 0.5–1 L stat of IV crystalloid
- Red cell transfusion to aim Hb >7 g/dL. Give 2 units of red cells if ongoing bleeding, haemodynamic instability, or Hb likely to drop below 7 g/dL. Target haemoglobin may be higher in patients with ischaemic heart disease or who remain unstable after initial resuscitation.

Reassess

- If advanced shock features or major blood loss (Box 8.2) on reassessment, resuscitate as per the adjacent column.
- Otherwise, continue with treatment and discuss with haematologist if any of the following:
 - Known coagulopathy/anticoagulant therapy
 - INR >1.4
 - Platelets <50 × 10⁹/L
 - Fibrinogen <1 g/L
 - ≥4 L IV fluid in any form

Advance shock features (Box 8.2) or ongoing major blood loss:

- Trigger major hemorrhage protocol, or as per local procedure.
- 1–2 L stat of IV crystalloid
- Nominate person to liaise with transfusion laboratory. Give group specific red cells as soon as available.
- Consider O negative blood if significant delay in cross-matching.

Reassess

- If ongoing advanced shock features or major blood loss, administer further red cells and consider fresh-frozen plasma, platelets and cryoprecipitate, depending on the local major haemorrhage protocol, the volume of blood lost, the patient's haematological parameters and Haematology advice.
- Discuss targeted therapy for patients' medications with Haematology, such as andexanet alfa for apixaban reversal, or prothrombin complex concentrate and vitamin K for warfarin.
- Consider terlipressin and initiate broad-spectrum antibiotics if variceal bleed suspected.
- Seek specialist input, such as ICU, GI, Surgery as appropriate and discuss ongoing haemostatic support with haematologist.

Box 8.2 Clinical features of hypovolaemic shock**Early**

HR >100 bpm
 Capillary refill time >2 secs
 Narrow pulse pressure
 Postural hypotension
 Urine output 20–30 mL/hr
 Pale, sweaty, anxious, thirsty

Advanced

Systolic BP <100 (or >40 mmHg drop from baseline)
 HR >120 bpm
 RR >20 breaths/min
 Cold, mottled peripheries
 Urine output < 20 mL/hr
 Confused, lethargic, low GCS score

1

**Resuscitation requirements assessed?
(Boxes 8.1 and 8.2)**

Resuscitation is the top priority. Use Boxes 8.1 and 8.2 as a framework for evaluating resuscitation needs, but tailor your assessment to

the individual. Monitoring trends of the parameters provides far more information than a single 'snapshot' assessment and is essential for evaluating response to treatment.

Do not rely exclusively on this framework. Be aware of different physiological responses

to haemorrhage: young patients may compensate until blood loss is profound; heart rate may be misleading in patients on rate-limiting medications (e.g., beta blockers, calcium channel blockers) or with fixed-rate pacemakers; and compensatory vasoconstriction may not occur in patients on vasodilators.

In patients who present with fresh, red haematemesis, the subsequent vomiting of 'coffee grounds' or passage of black tarry stools may simply be a manifestation of the original bleed, but further bright red haematemesis or haematochezia implies continued active bleeding.

2 Active bleeding, features of shock or suspected variceal bleed?

All patients with haematemesis accompanied by features of shock (Box 8.2) or evidence of ongoing bleeding should have an urgent UGIE after adequate resuscitation. Variceal bleeds have a high mortality (30–50%) and require urgent endoscopic identification and treatment to arrest the bleeding following resuscitation and correction of coagulopathy. Terlipressin and broad-spectrum antibiotics should be considered pre-UGIE in patients with a suspected variceal bleed. Assume variceal bleeding in any patient with known hepatic cirrhosis or clinical and biochemical features of chronic liver disease (Box 12.5).

3 Glasgow-Blatchford score ≥ 1 (Table 8.1)?

In the absence of continued active bleeding or haemodynamic compromise, use the Glasgow-Blatchford Score (Table 8.1) to assess the requirement for acute endoscopic intervention. A patient with a score of ≤ 1 has a minimal risk of needing an intervention such as transfusion, endoscopy or surgery. These patients can be considered for an early discharge and outpatient management (see below). Any score >1 has a higher risk for needing a medical intervention such as transfusion, endoscopy, or surgery and should be admitted for further assessment. Scores of 6

Table 8.1 Glasgow-Blatchford Score

Admission risk marker	Score component value
Blood urea (mmol/L)	
6.5–8.0	2
8.0–10.0	3
10.0–25	4
>25	6
Haemoglobin (g/dL) for men	
12.0–12.9	1
10.0–11.9	3
<10.0	6
Haemoglobin (g/dL) for women	
10.0–11.9	1
<10.0	6
Systolic blood pressure (mm Hg)	
100–109	1
90–99	2
<90	3
Other markers	
Pulse $\geq 100/\text{min}$	1
Melaena	1
Syncope	2
Hepatic disease	2
Cardiac failure	2
<i>A score of ≤ 1 is associated with minimal risk of requiring an intervention such as transfusion, endoscopy or surgery. These patients can be considered for an early discharge and outpatient management. Any score >1 has an increased risk of intervention. Scores ≥ 6 are associated with a $>50\%$ risk of needing an intervention.</i>	

or more are associated with a $>50\%$ risk of needing an intervention.

4 Presence of red-flag features?

Request an UGIE to exclude upper GI malignancy if the patient has any of the following red-flag features:

- Weight loss
- Anorexia or early satiety
- Dysphagia
- Epigastric mass
- Lymphadenopathy
- Jaundice
- Age ≥ 50 years

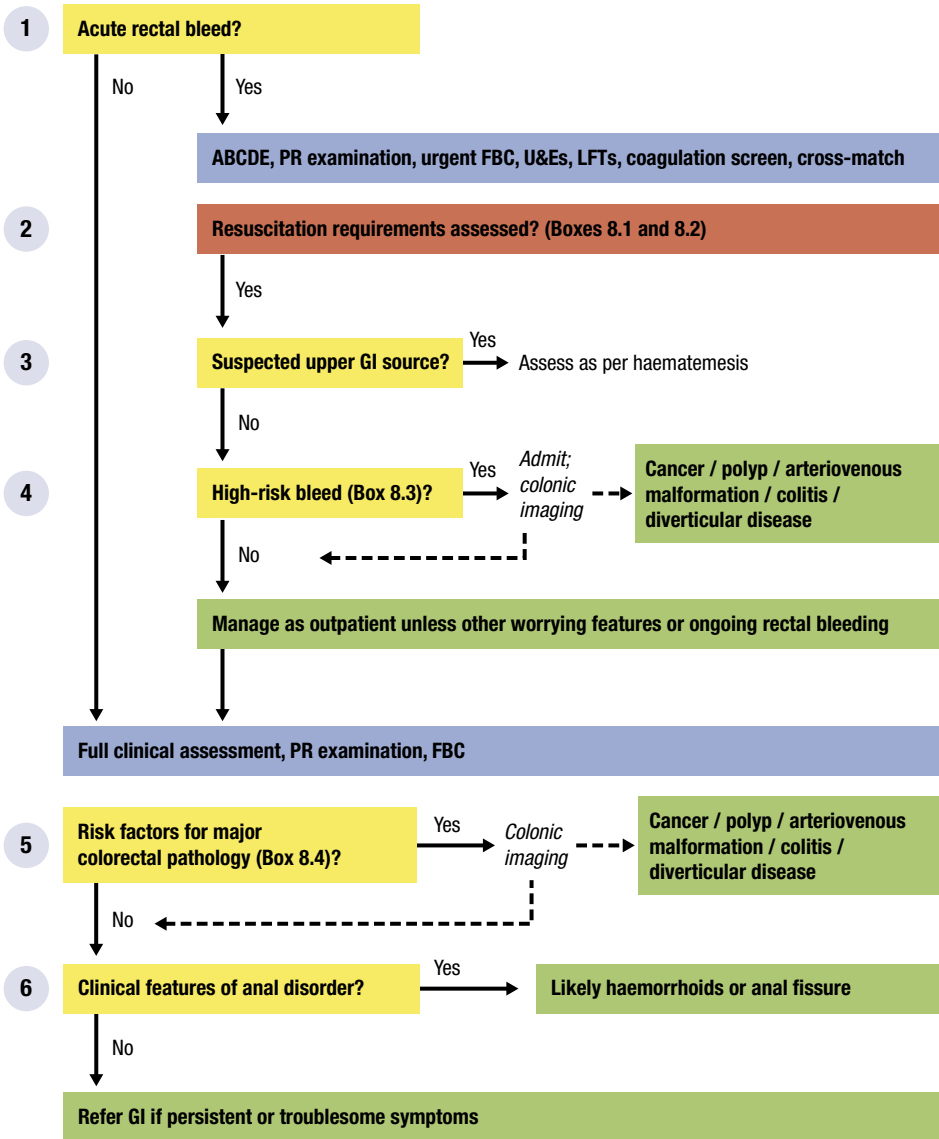
5

Consider Mallory-Weiss tear, gastritis or oesophagitis. Consider outpatient UGIE and arrange follow up to ensure resolution of symptoms

Patients who have no red flag features, no haemodynamic compromise and a Glasgow-Blatchford score of ≤ 1 can safely be discharged from hospital with outpatient follow-up. A working diagnosis may be suggested by the clinical history. With a characteristic history of Mallory-Weiss tear (haematemesis preceded

by forceful retching/vomiting), the diagnosis can often be made clinically, and treatment of the underlying cause may be indicated. In other cases, gastritis or oesophagitis may be the working diagnosis. Give an oral proton pump inhibitor and consider outpatient UGIE with a CLO test (for detection of *H pylori*). Arrange follow-up to ensure resolution of symptoms. If symptoms are unexplained or episodes of haematemesis are recurrent, arrange an UGIE.

Rectal bleeding: overview



Rectal bleeding: overview

Rectal bleeding: step-by-step assessment

1 Acute rectal bleed?

Regard any prolonged, profuse or ongoing episode of bright red/maroon PR blood loss within the previous 24 hours as acute rectal bleeding. Assess patients with melaena in the same way as those with haematemesis (see above).

Where the predominant problem is bloody diarrhoea, evaluate as described in [Chapter 9](#). In the absence of acute rectal bleeding, proceed to step 5.

2 Resuscitation requirements assessed? (Boxes 8.1 and 8.2)

Make adequate resuscitation your top priority. Evaluate and address resuscitation requirements as described in [Boxes 8.1 and 8.2](#).

3 Suspected upper GI source?

Assume, initially, that melaena reflects upper GI bleeding and assess as described for haematemesis. Brisk bleeding from the upper GI tract may present with a fresh red PR blood and implies severe, life-threatening haemorrhage. The diagnosis is straightforward if there is concomitant haematemesis but otherwise a high index of suspicion is essential. Proceed with a working diagnosis of lower GI bleeding in patients without haemodynamic instability, but suspect an upper GI source in those with features of hypovolaemic shock ([Box 8.2](#)).

If shock is present, seek senior GI input and consider UGIE as the first-line investigation following resuscitation, especially if there are any recent upper GI symptoms, such as epigastric pain, or known/suspected oesophageal varices. If the source is not identified on UGIE, then CT angiography and/or colonoscopy may be considered, informed by specialist opinion.

4 High-risk bleed (Box 8.3)?

Consider early discharge (with outpatient follow-up) if the patient has no evidence of ongoing bleeding and no high-risk features ([Box 8.3](#))—especially if there is evidence of a benign anal disorder. Do not discharge a patient without making a clear decision as to the likely cause of bleeding.

8**Box 8.3 Risk factors for adverse outcomes in acute lower GI bleeding**

- Any haemodynamic instability
- Initial haematocrit <35% or need for red blood cell transfusion
- Any visualized red blood PR
- INR >1.4 or current anticoagulant use (e.g., warfarin, novel oral anticoagulants, aspirin)
- Significant comorbidity, for example, cardiorespiratory, renal, hepatic impairment or current hospital inpatient
- Low serum albumin
- Raised WCC
- Age >60 years
- A prior history of bleeding from diverticulosis or angiodysplasia

Rectal bleeding: step-by-step assessment

Admit any patient with risk factors for uncontrolled or recurrent bleeding (Box 8.3). Once adequately resuscitated and clinically stable, arrange colonic imaging to establish the site and cause of bleeding. Most cases of rectal bleeding settle with conservative management. Be cautious about allowing the patient to mobilize to the toilet if they need to pass stool—this may represent another bleed and they risk collapse.

Seek urgent senior surgical input in any patient with continued active bleeding, ongoing haemodynamic instability despite resuscitation or high transfusion requirements. CT angiography is the first line intervention if there is clinical evidence of ongoing bleeding despite resuscitation. Colonoscopy is the most informative investigation and the intervention of choice if the bleeding has stopped as it facilitates a final diagnosis.

5 Risk factors for major colorectal pathology?

Refer the patient for urgent lower GI investigation (usually colonoscopy) to exclude colorectal cancer if any of the criteria in Box 8.4 are present.

Also consider lower GI endoscopic assessment if the patient has:

- Systemic/extraintestinal features of inflammatory bowel disease (Box 9.3)
- A history of pelvic radiotherapy

- Recent-onset, persistent bleeding when the patient is ≥ 50 years or has a strong family history of colorectal cancer, such as a first-degree relative < 45 years at diagnosis, or two or more first-degree relatives with the diagnosis.

Box 8.4 High-risk features for colorectal cancer in patients with rectal bleeding

- Rectal bleeding ≥ 4 weeks with change of bowel habit ($\downarrow$ or $\uparrow$ frequency or alternating bowel habit)
- Tenesmus
- Palpable rectal or abdominal mass
- Significant weight loss
- Previous excision of colorectal cancer or polyps
- History of inflammatory bowel disease
- Family history of colorectal cancer, polyps or inherited colorectal cancer syndromes
- Age > 50 years without anal symptoms, such as discomfort, itching, lumps, prolapse

6 Clinical features of anal disorder?

Perform careful rectal examination $\pm$ proctoscopy in all patients to look for an anal disorder. Do not forget to look for anal cancer. If benign anorectal pathology is identified, provide reassurance but refer the patient for outpatient colorectal evaluation if there is diagnostic doubt or symptoms persist despite conservative management.

Arrange further GI investigation $\pm$ referral in all patients without an obvious anorectal cause in whom bleeding persists or recurs.

Diarrhoea may be defined as the passage of ≥ 3 loose or watery stools/day. Patients often have difficulty describing their stools. The Bristol Stool Form Scale can be helpful (Fig. 9.1).

The differential diagnosis depends on symptom duration.

- **Acute diarrhoea** (<2 weeks): usually infectious and self-limiting, e.g., norovirus or rotavirus. Less common causes include drugs, inflammatory bowel disease and ischaemic colitis. Most patients do not present to healthcare services as the symptoms are transient. However, patients may seek medical attention if their symptoms are severe, there is associated systemic upset, they are becoming dehydrated or there are concerning features such as PR bleeding. This is more likely if symptoms are due to invasive pathogens, inflammatory bowel disease, or ischaemia. The differential diagnosis and investigation of diarrhoea in immunocompromised patients is markedly different, due to the extensive range of possible pathogens and severity of the illness. These patients are best discussed with Infectious Disease (ID) specialists early in their presentation.
- **Chronic/persistent diarrhoea** (>2 weeks): As symptoms become chronic, noninfective causes become pervasive in resource-rich environments. Diarrhoea may reflect colorectal cancer or inflammatory bowel disease, but the most frequent cause is irritable bowel syndrome. In resource-limited settings, infective causes remain dominant, including mycobacterial, chronic bacterial and parasitic causes.

Infectious diarrhoea

Infectious diarrhoea is due to transmission of viruses, bacteria, bacterial toxins or parasites,

usually through the faecal-oral route. Most cases are viral, self-limiting and the pathogen is rarely identified. Viruses and toxins predominantly affect the stomach and small bowel, causing large-volume, watery diarrhoea, vomiting, abdominal cramping, bloating and gas. Invasive intestinal pathogens cause inflammation of the large bowel and may cause bloody diarrhoea, often with severe abdominal cramps, passage of mucous and systemic upset (dysentery). Fever may be a feature. Pathogens include Shiga-toxin producing *Escherichia coli* (e.g., O157), *Shigella*, *Salmonella*, *Campylobacter*, enteric viruses (e.g., cytomegalovirus, adenovirus) and cytotoxic organisms such as *Clostridium difficile*. *C. difficile* infection is an important cause of hospital-acquired diarrhoea, risk factors for which include broad-spectrum antibiotic therapy, immunocompromise, and advanced age. *C. difficile* ranges from mild diarrhoeal illness to life-threatening pseudomembranous colitis.

Diarrhoea that persists for >10 days is unlikely to be infective, but consider protozoal infections (e.g., giardiasis, amoebiasis or *Cryptosporidium* infection) in patients who are immunocompromised or have recently travelled to the tropics.

Irritable bowel syndrome

Irritable bowel syndrome (IBS) is the most common cause of chronic diarrhoea. The predominant bowel habit may alternate between diarrhoea and constipation. Diagnosis is based on typical clinical features (Box 9.1) and the absence of apparent organic disease. Symptoms tend to follow a relapsing and remitting course, often exacerbated by psychosocial stress. There are no absolutes when considering investigations to rule out organic disease if IBS is suspected. While many female patients report cyclical features

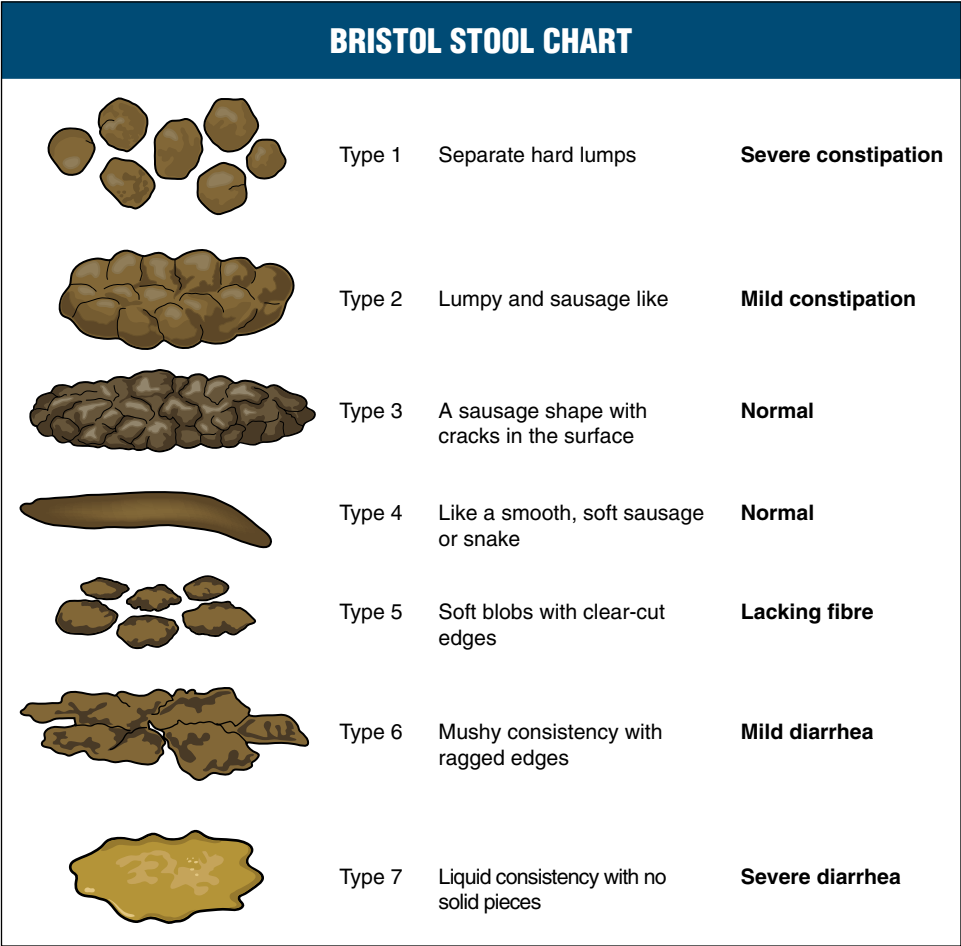


Figure 9.1 Bristol stool form scale. From Arzu Ilce. Fecal incontinence: causes, management, and outcome. Chapter 2: Faecal incontinence. IntechOpen, 2014. <http://dx.doi.org/10.5772/57502>.

to their symptoms correlating to menstruation, it is important to rule out endometriosis as a cause for the patient’s GI features. Have a low threshold for detailed investigation in patients with a known family history of ovarian or colorectal cancer.

Drugs

Many drugs, including numerous over-the-counter preparations, cause diarrhoea (Box 9.2).

Colorectal cancer

Colorectal cancer may present with diarrhoea, especially if left-sided/distal. Suggestive features include weight loss, rectal bleeding, a palpable mass or iron deficiency anaemia, but the absence of these features does not exclude malignancy. The diagnosis must be considered in any patient >45 years with new-onset, persistent diarrhoea and is usually confirmed by colonoscopy with

Box 9.1 Rome IV criteria for diagnosis of irritable bowel syndrome

Irritable bowel syndrome diagnostic criteria:^a

Recurrent abdominal pain on average at least 1 day/week in the last 3 months, associated with two or more of the following criteria:

- Related to defecation
- Associated with a change in frequency of stool
- Associated with a change in form (appearance) of stool

^aCriteria fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis.

Reprinted with permission from the Rome Foundation; all rights reserved.

Box 9.2 Drugs that frequently cause diarrhoea

- Laxatives (including occult laxative misuse)
- Antibiotics (especially macrolides)
- Alcohol (especially chronic alcohol excess)
- NSAIDs
- Metformin
- Colchicine
- Orlistat (steatorrhoea)
- Proton pump inhibitors
- SSRIs
- Nicorandil
- Digoxin
- Cytotoxic and immunosuppressive agents
- Cancer immunotherapies
- Magnesium replacement and antacids (due to magnesium content)

biopsy. Many countries now conduct colorectal cancer screening programmes, which has led to more cases being diagnosed at an asymptomatic stage; diagnosis following presentation with symptoms is associated with a worse prognosis.

Inflammatory bowel disease

Ulcerative colitis (UC) is an inflammatory disorder of the colonic mucosa confined to the large bowel. It typically presents with bloody diarrhoea with mucous and cramping lower abdominal pain $\pm$ tenesmus. Systemic features that occur during acute attacks include fever, anorexia and weight loss.

Crohn's disease causes transmural granulomatous inflammation affecting any part of the alimentary tract. It may present with large

Box 9.3 Extraintestinal features of inflammatory bowel disease

- General: fever, malaise, weight loss
- Eyes: conjunctivitis, episcleritis, iritis
- Joints: arthralgia of large joints, seronegative spondyloarthritis
- Skin: mouth ulcers, erythema nodosum, pyoderma gangrenosum
- Liver: fatty liver, gallstones, sclerosing cholangitis, cholangiocarcinoma (UC)

bowel symptoms, like those of UC, or with small bowel symptoms, such as watery, non-bloody diarrhoea accompanied by abdominal pain and weight loss. Both disorders are associated with a range of extraintestinal features (Box 9.3).

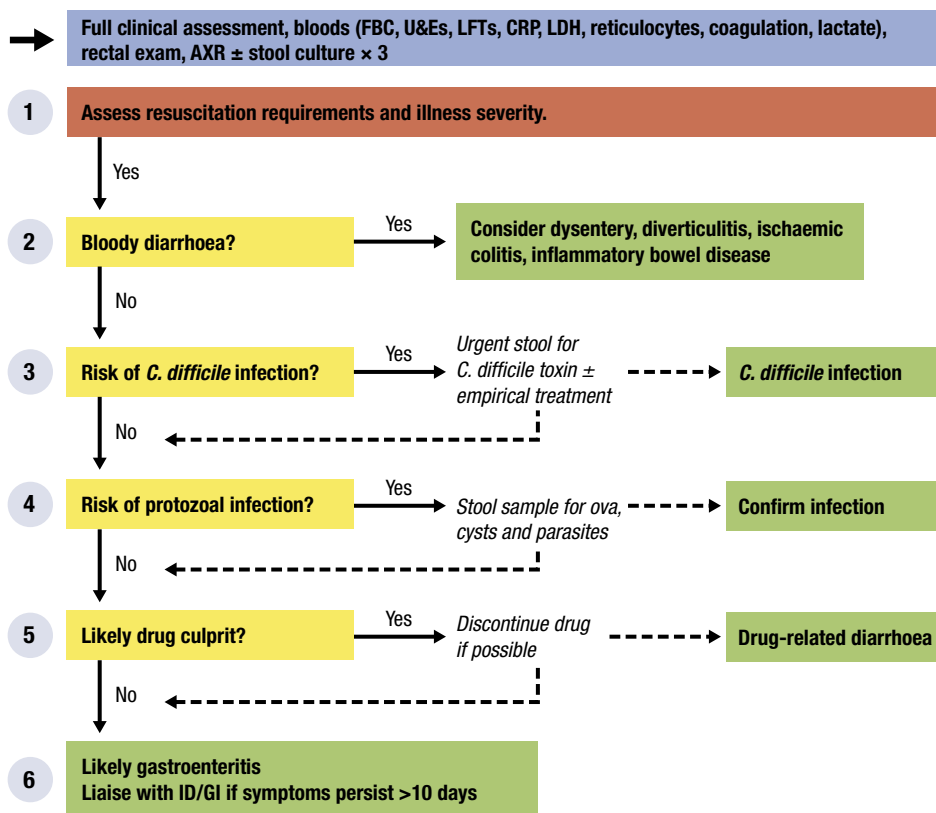
Microscopic colitis is an inflammatory condition affecting the mucosa of the colon. It is characterized by watery, nonbloody diarrhoea, predominantly in the female population. Macroscopically, the mucosa appears normal and diagnosis is made on histological appearances. While there are a number of autoimmune disorders associated with microscopic colitis, there are no hallmark extraintestinal features.

Disorders causing malabsorption

Fat malabsorption causes pale, greasy, offensive stools that float and are difficult to flush (steatorrhoea). Other features of malabsorption include undigested foodstuffs in stool, weight loss, bloating and nutritional deficiencies. The usual underlying cause is small bowel disease, for example, coeliac disease, Crohn's disease, tropical sprue, bacterial overgrowth, lymphoma, small bowel resection or pancreatic insufficiency.

Other causes

- Diverticulitis (Chapter 7)
- Ischaemic colitis (Chapter 8)
- Hyperthyroidism, autonomic neuropathy, Addison's disease
- Carcinoid tumour, gastrinoma, VIPoma
- Amyloid
- Severe constipation with overflow



Acute diarrhoea: overview

1 Assess resuscitation requirements and illness severity

Step 1 Are there features of shock?

Look for ↑HR, ↓BP (a late sign) or evidence of tissue hypoperfusion (Box 2.1). If present, provide IV fluid resuscitation and reassess. Although hypovolaemia from GI losses is the likeliest cause of shock, also consider intra-abdominal sepsis, and adrenal insufficiency (diarrhoea is a common feature of acute adrenal crisis).

Step 2 Is there acute kidney injury?

Hypovolaemia may result in severe 'prerenal' acute kidney injury (AKI, Chapter 29), especially if compounded by antihypertensive or nephrotoxic medication, such as diuretics, ACE inhibitors, NSAIDs. Patients with AKI require IV rehydration with close monitoring of fluid balance, urine output and U+Es. If AKI occurs in the context of bloody diarrhoea,

look for features of haemolytic uraemic syndrome, such as ↓Hb, ↓platelets, ↑bilirubin/LDH/reticulocytes. This is a hallmark and sign of disease severity in *E coli* O157 infection.

Step 3 Does the patient otherwise require IV fluids/hospital admission?

Patients with clinical evidence of dehydration (thirst, dry mucous membranes, ↓skin turgor) and concomitant vomiting, or those who are unable to match oral intake to ongoing losses, require IV fluids.

Other features that may indicate a need for hospital admission include:

- Fever
- ↑WBC
- Bloody diarrhoea
- Abdominal tenderness/guarding/rigidity
- Frail, elderly or immunocompromised patient
- Significant comorbidity, such as heart/renal/hepatic failure

2 Bloody diarrhoea?

The frequent passage of bloody stools suggests either:

- Dysentery, that is, infection with invasive organisms (e.g., *Campylobacter*, *Shigella* or *Amoeba*) or cytotoxin-producing organisms (e.g., *C difficile*, *E coli* O157) OR
- Noninfectious colitis, for example, ischaemic colitis or inflammatory bowel disease

It is often difficult to differentiate dysentery from early presentations of inflammatory bowel disease: both may be accompanied by abdominal pain, tenesmus, mucus, constitutional upset and a systemic inflammatory response (Box 7.1).

- Send FBC, U+Es, LFTs, CRP, LDH, reticulocytes, venous blood gas and three stool samples in all patients, plus blood cultures if fever is present.
- Test stool for *C difficile* toxin (CDT) if the patient has risk factors (see below), severe systemic upset or 1WBC.
- Request an abdominal X-ray to look for colonic dilatation (Fig. 7.5B).
- Liaise with the ID team and request analysis of stool for ova, cysts and parasites if recent foreign travel or immunocompromised.

If the duration of symptoms is ≤7 days, suspect an infectious cause unless there are specific pointers to an alternative diagnosis. Consider ischaemic colitis if bloody diarrhoea was preceded by sudden onset of left-sided lower abdominal pain or in any patient >50 years with known atherosclerotic disease or a source of systemic embolism, such as atrial fibrillation.

Refer any patient with known inflammatory bowel disease, extraintestinal manifestations (Box 9.3), previous similar episodes or symptoms >7 days for specialist GI evaluation.

Seek an urgent surgical review if there is evidence of peritonism or toxic megacolon, or if you suspect ischaemic colitis or diverticulitis.

3 Risk of *C difficile* infection?

Send stool for CDT in any patient who lives in an institution (e.g., nursing home), has recently been hospitalized, has received antibiotics within the last 3 months or is >65 years. In high-risk patients, send ≥3 samples before ruling out the diagnosis. If *C difficile* is confirmed,

stop contributing antibiotics and proton pump inhibitors if possible, assess severity at least daily (Box 9.4) and treat as per local guidance.

4 Risk of protozoal infection?

Send three stool samples on consecutive days for ova, cysts and parasites in patients with a history of recent foreign travel or when there is known or suspected immunocompromise (e.g., chemotherapy, HIV).

5 Likely drug culprit?

Suspect drug-related diarrhoea if the onset of symptoms corresponds with initiation or 1 dose of a drug, especially those listed in Box 9.2. Seek an alternative explanation if diarrhoea does not resolve on drug discontinuation.

6 Likely gastroenteritis. Liaise with ID/GI if symptoms persist >10 days

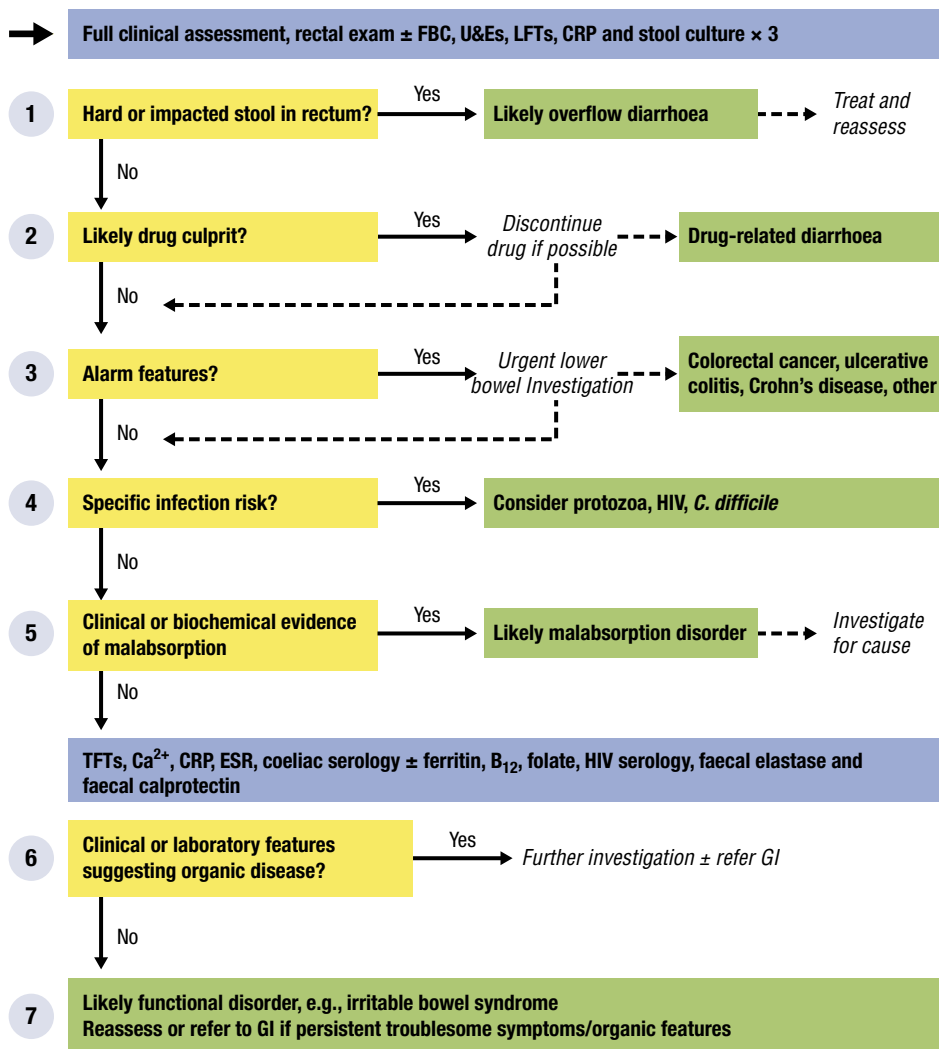
Most cases are self-limiting viral or toxin-mediated infections and do not require further investigation or antimicrobial treatment. If symptoms persist >10 days, seek specialist advice and consider further assessment as for chronic/persistent diarrhoea.

Box 9.4 Assessing the severity of *Clostridium difficile* infection

Mild	WBC not elevated <3 episodes of loose stools/day
Moderate	1WBC (but <15 × 10 ⁹ /L) 3–5 loose stools/day
Severe	WBC 15 × 10 ⁹ /L or Serum creatinine >50% above baseline or Temp >38.5°C or Evidence of severe colitis (abdominal or radiological signs)
Life-threatening	Signs of shock Partial or complete ileus Toxic megacolon or CT evidence of severe disease

Adapted from NICE 2021 (*Clostridioides difficile* infection: antimicrobial prescribing). Available from www.nice.org.uk/guidance/ng199. All rights reserved. Subject to Notice of rights. NICE guidance is prepared for the National Health Service in England. It is subject to regular review and updating and may be withdrawn. NICE accepts no responsibility for the use of its content in this product/publication.

Chronic/persistent diarrhoea: overview



Chronic/persistent diarrhoea: overview

1 Hard or impacted stool in rectum?

Have a high index of suspicion for overflow diarrhoea in frail, immobile or confused elderly patients. Always do a rectal examination. If the stool is hard or impacted, treat with faecal softeners and laxatives, then reassess. If the rectal examination is normal, overflow diarrhoea is unlikely (90% of impactions are rectal) but consider an abdominal X-ray if there is strong clinical suspicion.

2 Likely drug culprit?

Look for a temporal relationship between potential drug culprits (Box 9.2) and the onset of diarrhoea. Consider trialling discontinuation where feasible, but change only one agent at any time and restart the drug if symptoms continue unaltered. Always ask about alcohol excess.

3 Alarm features?

Expedite lower bowel investigation to exclude colorectal cancer/inflammatory bowel disease

Chronic/persistent diarrhoea: step-by-step assessment

if the patient has persistent diarrhoea associated with any of the following:

- Age >45 years
- Rectal bleeding or melaena
- Weight loss
- Iron-deficiency anaemia
- Pain or diarrhoea overnight
- Progressive abdominal pain
- Fever, raised inflammatory markers or other systemic symptoms
- Palpable rectal mass
- Family history of inflammatory bowel disease or colorectal cancer

Consider flexible sigmoidoscopy as a first-line investigation in patients <45 years or colonoscopy if >45 years.

4 Specific infection risk?

Exclude protozoal infection in patients with a history of travel to the tropics (e.g., giardiasis, amoebiasis) or with known HIV infection/other immunocompromise (e.g., cryptosporidiosis). Send three fresh stool samples for examination for ova, cysts and parasites. Liaise with the ID team if stool samples are negative but there is ongoing clinical suspicion of infection. Test for HIV in all patients with chronic diarrhoea.

C difficile diarrhoea may be chronic or relapsing—send stool to test for CDT in any patient with risk factors (see above).

5 Clinical or biochemical evidence of malabsorption?

Steatorrhoea signifies fat malabsorption, but its absence does not exclude a malabsorptive disorder. Screen for nutritional deficiencies (e.g., iron, B₁₂, folate, Ca²⁺, Mg²⁺, PO₄³⁻, albumin) in any patient with suspected malabsorption. If steatorrhoea is present, ensure that the patient is not taking orlistat (available over the counter in the UK), and check coeliac serology and faecal elastase (↓in

pancreatic insufficiency). If ↓faecal elastase or a strong suspicion of pancreatic disease (e.g., suggestive symptoms, history of acute pancreatitis, cystic fibrosis), consider pancreatic imaging (CT/MRCP). If coeliac serology is positive, then arrange confirmatory testing for coeliac disease with endoscopy and duodenal biopsy. Note that coeliac serology and endoscopy with biopsy will detect coeliac disease only if the patient has recently been exposed to gluten; patients should continue to eat gluten-containing food during investigation.

If there is no evidence of pancreatic insufficiency or coeliac disease, but suspicion of malabsorption remains high (e.g., large-volume non-bloody stool, previous gastric/small bowel surgery or evidence of nutritional deficiencies), refer to GI for further small bowel investigation, such as duodenal biopsy or MRI.

6 Clinical or laboratory features suggesting organic disease?

Exclude inflammatory bowel disease if there is a positive family history, evidence of systemic inflammation, extra-intestinal manifestations of inflammatory bowel disease (Box 9.3) or positive faecal calprotectin. Refer to GI for further evaluation if any of the above features are present or there are other findings that suggest organic disease, such as weight loss, anorexia, painless diarrhoea, prominent nocturnal symptoms or recent onset of symptoms in a patient >45 years.

7 Likely functional disorder

If none of the above is present, a functional cause, such as irritable bowel syndrome, is likely, particularly when typical symptoms are present (Box 9.1) and baseline investigations are negative. Provide reassurance, explanation and institute a management plan.

10 Dysphagia

Dysphagia is the symptom of difficult or abnormal swallowing. It is an alarm ('red flag') symptom and therefore requires prompt assessment to exclude serious pathology. With the exception of acute presentations with a concomitant sore throat, oesophageal investigation is always necessary to rule out mechanical obstruction and, in particular, malignancy. History is critical to informing the differential diagnosis and can help determine whether the condition causing dysphagia is a structural or motility disorder. Structural disorders typically cause dysphagia for solids; motility disorders may cause dysphagia for solids and liquids.

Oropharyngeal dysphagia

Bulbar palsy

Lower motor neuron palsies of cranial nerves IX–XII result in weakness of the tongue and muscles of swallowing. The tongue is flaccid with fasciculation, often with a change in voice (Table 10.1). Causes include motor neuron disease (MND), Guillain-Barré syndrome, syringobulbia and brainstem tumour/infarction.

Pseudobulbar (corticobulbar) palsy

Bilateral upper motor neuron lesions of cranial nerves IX–XII cause dysphagia with a small, contracted and slowly moving tongue, weakness or paralysis of the pharyngeal muscles and a brisk jaw jerk. There may be associated speech disturbance and emotional lability (Table 10.1). It is more common than bulbar palsy, and causes include cerebrovascular disease, demyelination, MND and central pontine myelinolysis.

Myasthenia gravis

Myasthenia gravis causes fatigability of oropharyngeal muscles, resulting in increasing swallowing difficulty after the first few mouthfuls of

food/fluid. Dysphagia may occur before other features of myasthenia are readily apparent.

Parkinson's disease, stroke and myopathies

These conditions frequently cause swallowing difficulty, but other features are usually more prominent.

Pharyngeal pouch

A pharyngeal pouch is formed by posterior herniation of the pharyngeal mucosa between the thyropharyngeus and cricopharyngeus muscles. It is usually found in elderly patients. In addition to dysphagia, there is classically regurgitation of undigested food, halitosis, the feeling of a lump in the neck and gurgling after swallowing liquids.

Oesophageal dysphagia (structural)

Food bolus obstruction

A food bolus obstruction causes sudden onset of complete dysphagia, often with an inability to swallow even saliva. The diagnosis is usually obvious from the history, but it may be the first manifestation of an underlying stricture.

Malignant stricture

Patients with oesophageal cancer typically present with progressive, painless dysphagia to solid foods. Weight loss may be marked, especially if presentation is delayed. Cachexia and lymphadenopathy are suggestive, but physical signs are typically absent. Risk factors include alcohol, smoking and known gastro-oesophageal reflux disease (GORD)/Barrett's oesophagus.

Benign stricture

Most commonly, benign stricture is due to gastro-oesophageal reflux, especially in elderly patients. Rarer causes include strictures weeks or months after ingestion of caustic substances,

Table 10.1 Comparison of signs in bulbar and pseudobulbar palsy

	Bulbar	Pseudobulbar
Speech	Quiet, nasal tone; difficulty forming consonants (especially 'R'); may become slurred	Slow deliberate speech
Tongue	Weak, wasted with fasciculation, folded	Small, stiff, cannot be protruded
Jaw jerk	Normal or absent	Brisk
Gag reflex	Absent	Present
Emotions	Normal	Labile (e.g., uncontrollable laughing/crying)

oesophageal webs (seen in iron deficiency), Schatzki rings (benign idiopathic strictures of the lower oesophagus), eosinophilic oesophagitis or, rarely, benign oesophageal tumours.

Hiatus hernia

Hiatus hernia with an associated intrathoracic stomach can present with dysphagia and vomiting. GORD manifests only when the lower oesophageal sphincter becomes incompetent.

Extrinsic compression

Lung cancer, thyroid goitre, mediastinal nodes, an enlarged left atrium or a thoracic aortic aneurysm may produce dysphagia by compressing the oesophagus.

Oesophageal dysphagia (dysmotility)

Achalasia

This is an uncommon disorder characterized by loss of peristalsis in the distal oesophagus and impaired relaxation of the lower oesophageal sphincter. Dysphagia is of slow onset (often

years), occurs for liquids and solids and may initially be intermittent. Dysphagia for liquids is the most prominent symptom. Retrosternal discomfort and regurgitation are common.

Scleroderma

Oesophageal involvement is seen in ~90% of cases of scleroderma; replacement of muscle with fibrous tissue results in incompetence of the lower oesophageal sphincter, impaired peristalsis and severe gastro-oesophageal reflux. Other features of scleroderma may be apparent, such as calcinosis, Raynaud's disease and telangiectasia.

Chagas disease

Chagas disease (a tropical parasitic disease caused by *Trypanosoma cruzi*) can present with achalasia; consider it in patients originating from endemic areas in Central and South America and confirm with serological testing.

Diffuse oesophageal spasm

This may cause transient episodes of dysphagia, although episodic chest pain that mimics angina is usually the predominant symptom.

Other, rarer causes

Inadequate saliva

Inadequate saliva production, which can be caused by anticholinergic side effects or connective tissue diseases such as Sjogren syndrome, may lead to problems with forming a manageable bolus and the sensation of dysphagia.

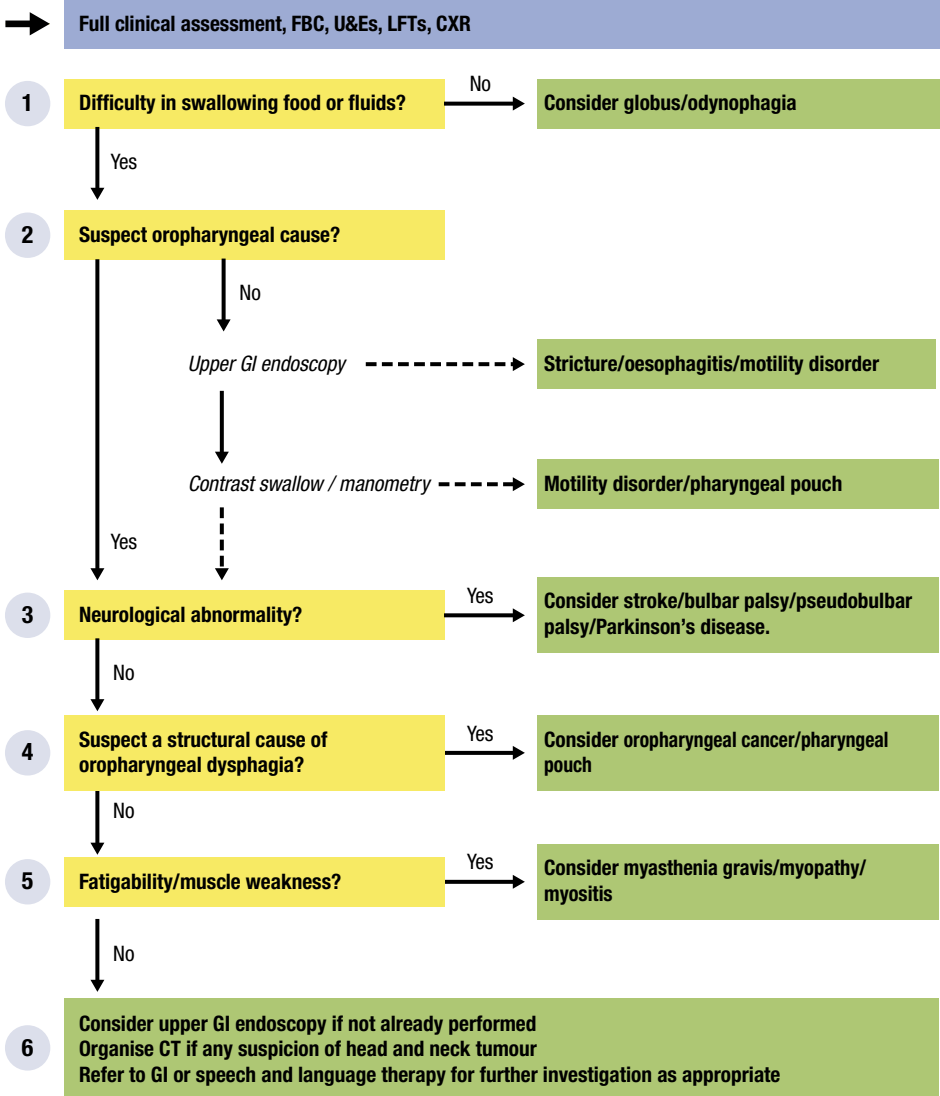
Globus

Globus is a functional disorder defined as the sensation of a lump or foreign body in the throat. It is a diagnosis of exclusion and requires that all of the Rome IV diagnostic criteria (Box 10.1) be met.

Box 10.1 Rome IV criteria for diagnosis of globus

To meet the diagnosis of globus, all of the following criteria must be fulfilled for a duration of at least 3 months, with symptom onset at least 6 months prior to diagnosis and symptom frequency of at least once per week.

1. Persistent or intermittent, nonpainful sensation of a lump or foreign body in the throat with no structural lesion identified on physical examination, laryngoscopy, or endoscopy
2. Occurrence of the sensation between meals
3. Absence of dysphagia or odynophagia
4. Absence of a gastric inlet patch in the proximal oesophagus
5. Absence of evidence that gastroesophageal reflux or eosinophilic oesophagitis is the cause of the symptom
6. Absence of major oesophageal motor disorders, such as achalasia/esophagogastric junction outflow obstruction, diffuse oesophageal spasm, jackhammer oesophagus, absent peristalsis



1 Difficulty in swallowing foods or fluids?

Features of true dysphagia include difficulty initiating swallowing or the sensation of food 'sticking' after swallowing. Globus is the sensation of a lump in the throat; it is unrelated to swallowing and often associated with anxiety or strong emotion. Odynophagia is pain on swallowing and suggests oesophageal inflammation or ulceration.

2 Suspect oropharyngeal cause?

Suspect oropharyngeal dysphagia if the patient reports one or more of the following:

- Evidence of oral dysfunction: drooling/spillage of food from the mouth, dysarthria.
- Immediate sensation of a bolus 'catching' in the neck. Patients may identify their neck as the point of their symptoms rather than retrosternally, which is associated more with oesophageal dysphagia.
- Difficulty initiating swallow—repeated attempts are required to clear the bolus.
- Choking/coughing/aspiration on swallowing.

In patients in whom oropharyngeal dysphagia is suspected, progress to Step 3 for further investigation.

If none of these features is present, investigate for an oesophageal cause. Arrange an UGIE to look for a structural cause, especially oesophageal cancer. upper GI endoscopy (UGIE) may also reveal features of certain motility disorders (e.g., achalasia) but can be normal, especially in scleroderma or diffuse oesophageal spasm. Note that a contrast barium swallow test may be first-line investigation if the patient has a history of caustic ingestion, radiotherapy or laryngeal or oesophageal cancer—discuss with a GI specialist in the first instance.

If no cause is identified on UGIE, consult a GI specialist, as further investigation for a motility disorder (e.g., contrast swallow or oesophageal manometry) may be required. If all GI investigations are reassuring, continue to evaluate for a possible oropharyngeal cause and progress to Step 3

3 Neurological abnormality?

If the history suggests an oropharyngeal aetiology for dysphagia, neurological evaluation

is essential. Although swallowing difficulty is common in stroke, Parkinson's disease and multiple sclerosis, it is very rarely the primary presenting problem. However, dysphagia may be the principal complaint in bulbar palsy and pseudobulbar palsy, so look carefully for the relevant clinical features (Table 10.1); if they are present, look for other features of MND and perform neuroimaging to exclude vascular or structural brainstem disease. Discuss with a neurologist all patients in whom there is concern about a neurological abnormality causing dysphagia.

4 Suspect structural cause of oropharyngeal dysphagia?

Exclude oropharyngeal cancer if the patient has an oropharyngeal pattern of dysphagia (Step 2) without evidence of an underlying neurological disorder. Risk factors for oropharyngeal cancer include smoking, alcohol and human papillomavirus infection. A palpable neck lump, halitosis, cough, hoarseness, sore throat, ear pain (referred pain) and weight loss are some of the clinical features that may be associated. If suspected, refer to an ENT specialist for investigation.

Investigate for a pharyngeal pouch (e.g., contrast swallow or nasoendoscopy) if there is associated regurgitation or a lump in the neck that appears after eating $\pm$ halitosis (from food lodged in the pouch), weight loss, aspiration or chronic cough.

5 Fatigability/muscle weakness?

Myasthenia gravis is rare and easily missed. The dysphagia typically presents with increasing difficulty in swallowing after the first few mouthfuls and difficulty in chewing. Ask specifically about/examine for:

- Fatigable weakness of muscles: initial strength is normal but there is a rapid decline with activity
- Weak voice with prolonged speaking
- Diplopia
- Ptosis
- Muscle fatigability (Ask the patient to count to 50 or hold the arms above the head.)

Refer to a neurologist for definitive testing if you suspect the diagnosis.

6

Consider further investigation and referral to GI or speech and language therapy

All patients should be discussed with a GI specialist. If not already done, arrange UGIE unless there is clear evidence of neurological or neuromuscular dysfunction (where referral to neurology would be essential). Request a CT if you suspect cancer of the head and neck.

Once significant pathology of the oropharynx and oesophagus has been excluded, consider referral to speech and language therapy team for a detailed swallow evaluation with video fluoroscopy. This will confirm the presence of oropharyngeal dysfunction and help to clarify the mechanism.

Nausea is the unpleasant sensation of feeling as though you will vomit. It can be associated with vomiting (the forceful expulsion of gastric contents) or dyspepsia (symptoms of indigestion). Vomiting should be differentiated from regurgitation, which is the appearance of gastric contents without any effort. Acute (≤ 10 days) nausea and vomiting (N&V) should be assessed separately from chronic presentations.

This chapter focuses on N&V as a principal presenting complaint. See [Chapter 8](#) if GI bleeding is a feature.

Gastroenteritis

Gastroenteritis is the most common cause of acute N&V. The aetiology is usually viral, e.g., norovirus, rotavirus, adenovirus. It may affect any age group but has a higher incidence and associated harm in children and the elderly. Gastroenteritis may also be caused by bacteria or parasites. The sudden onset of N&V is likely due to the ingestion of a toxin such as *Staphylococcus aureus* enterotoxin or *Bacillus cereus* emetic toxin; vomiting tends to develop within 1–6 hours of ingestion of poorly prepared/cooked foodstuffs. Enteric viruses (e.g., norovirus, rotavirus) are transmitted via the faeco-oral route and manifest slightly later. Parasitic infections causing GI upset include nematodes and are associated with ingestion of raw fish.

A detailed history may reveal known contacts with similar symptoms or a history of travel, eating out or eating unusual foods recently. Accompanying features may include fever, abdominal cramping and diarrhoea.

GI obstruction

Mechanical obstruction of the GI tract (e.g., due to a tumour, adhesions or incarcerated hernia) may cause vomiting, usually accompanied by colicky abdominal pain ([Chapter 7](#))

and absolute constipation (absence of flatus) $\pm$ abdominal distension. The characteristics of the vomitus may suggest the location of obstruction: gastric juices—gastric outlet obstruction; bilious material—small bowel obstruction; feculent material—distal obstruction (or colo-enteric/cologastric fistula).

Gastroparesis

Gastroparesis is a disorder which causes delayed gastric emptying and N&V as a result of reduced peristalsis of the stomach. It is frequently associated with autonomic neuropathy and so is common in patients with diabetes. It can also be associated with other conditions such as scleroderma and can occur in patients who have had previous gastric surgery. In a significant proportion of patients, it is idiopathic.

Functional nausea and vomiting disorders

Functional nausea and vomiting disorders are diagnoses of exclusion, but there are symptoms to be alert for. Cyclical vomiting syndrome is characterized by repeated, sudden-onset, severe N&V with no identified cause; discrete episodes of N&V typically last <1 week. Patients with cannabinoid hyperemesis syndrome may report symptoms similar to cyclical vomiting. The syndrome is associated with prolonged cannabis use; symptoms may be alleviated by prolonged warm baths and showers. Functional nausea and vomiting disorders have specific criteria that must be met for at least 3 months, with symptom onset at least 6 months before diagnosis.

Other GI tract disease

Peptic ulcer disease may present with N&V, typically associated with epigastric discomfort or a history of dyspeptic symptoms. N&V may

Differential diagnosis**Box 11.1 Common toxic/therapeutic causes of nausea and vomiting**

Alcohol	Colchicine
Antiarrhythmics	Digoxin
Anticonvulsants	NSAIDs/Aspirin
Antibiotics (especially erythromycin and tetracyclines)	Oestrogen and progestogen-containing drugs
Anti-Parkinsonian treatments	Opioids
Cannabinoids	Psychotropic medications
Chemotherapy agents	Theophylline

be a prominent feature of acute inflammatory abdominal pathology (e.g., acute pancreatitis/appendicitis/cholecystitis) but abdominal pain is almost always the principal complaint ([Chapter 7](#)). Severe constipation may lead to N&V, particularly in frail elderly patients.

Medications

Numerous medications or toxins can result in N&V; some of the more common are listed in [Box 11.1](#). Alcohol misuse is a very frequent cause of both acute and chronic N&V.

Hypercalcaemia

N&V may be the main presenting symptom in hypercalcaemia, especially since other features tend to be nonspecific, e.g., fatigue, depression, constipation, abdominal pain, anorexia. Confusion and renal impairment may complicate severe hypercalcaemia. Patients presenting with hypercalcaemia require investigations to identify the underlying cause, which may be life-threatening. The main diagnoses to consider are malignancy (myeloma, bone metastases, ectopic secretion of parathyroid hormone-related peptide from tumour) and primary hyperparathyroidism. Rarer causes include sarcoidosis, adrenal insufficiency, lithium toxicity and tertiary hyperparathyroidism.

Box 11.2 Clinical/biochemical features often present in patients with adrenal insufficiency**Symptoms (often insidious)**

- Lethargy, fatigue, low mood
- Anorexia, weight loss
- Nausea and vomiting
- Postural hypotension
- Abdominal pain
- Diarrhoea/constipation

Physical signs

- Pigmentation (palmar creases, mucous membranes)
- Vitiligo
- Postural hypotension
- In Addisonian crisis: shock (low BP, reduced tissue perfusion), pyrexia, coma

Blood

- Hyponatraemia
- Hyperkalaemia
- Hypoglycaemia (fasting or spontaneous)
- Hypercalcaemia
- Anaemia
- Normal anion gap metabolic acidosis (mild)

Other

- History of other autoimmune disorders

Other metabolic disturbance

Chronic renal failure with uraemia tends to present nonspecifically, and nausea ± vomiting is often a prominent feature. Conversely, severe N&V is an important cause of acute kidney injury due to hypovolaemia. The assessment of acute kidney injury is discussed in [Chapter 29](#).

Diabetic ketoacidosis and, to a lesser extent, hyperosmolar hyperglycaemic state (HHS), may present with N&V. Suggestive features include a history of polyuria and/or polydipsia and a background of diabetes (particularly type 1).

GI upset, including N&V, is also common in adrenal insufficiency, including acute adrenal crisis. Hypotension and illness severity are

usually disproportionate to the degree of vomiting. Other suggestive features are shown in [Box 11.2](#).

CNS causes

Raised intracranial pressure (ICP) may produce vomiting, often without nausea. The vomiting can be profound. When raised ICP is chronic (e.g., space-occupying lesion), vomiting classically occurs shortly after waking in the morning and is associated with headache (worse on lying, bending or straining) and, possibly, papilloedema. In acute causes (e.g., intracerebral haemorrhage), there may be sudden-onset headache with vomiting followed by progressive neurological symptoms, including reduced consciousness level. Nausea, with or without vomiting, occurs during episodes of migraine in $\approx 80\%$ of patients and is often severe; however, headache $\pm$ aura is usually the dominant symptom ([Chapter 13](#)).

N&V accompanied by vertigo suggests vestibular or brainstem pathology ([Chapter 17](#)). Vomiting may also occur in CNS infection

(e.g., meningitis, encephalitis) but usually not as the main presenting feature.

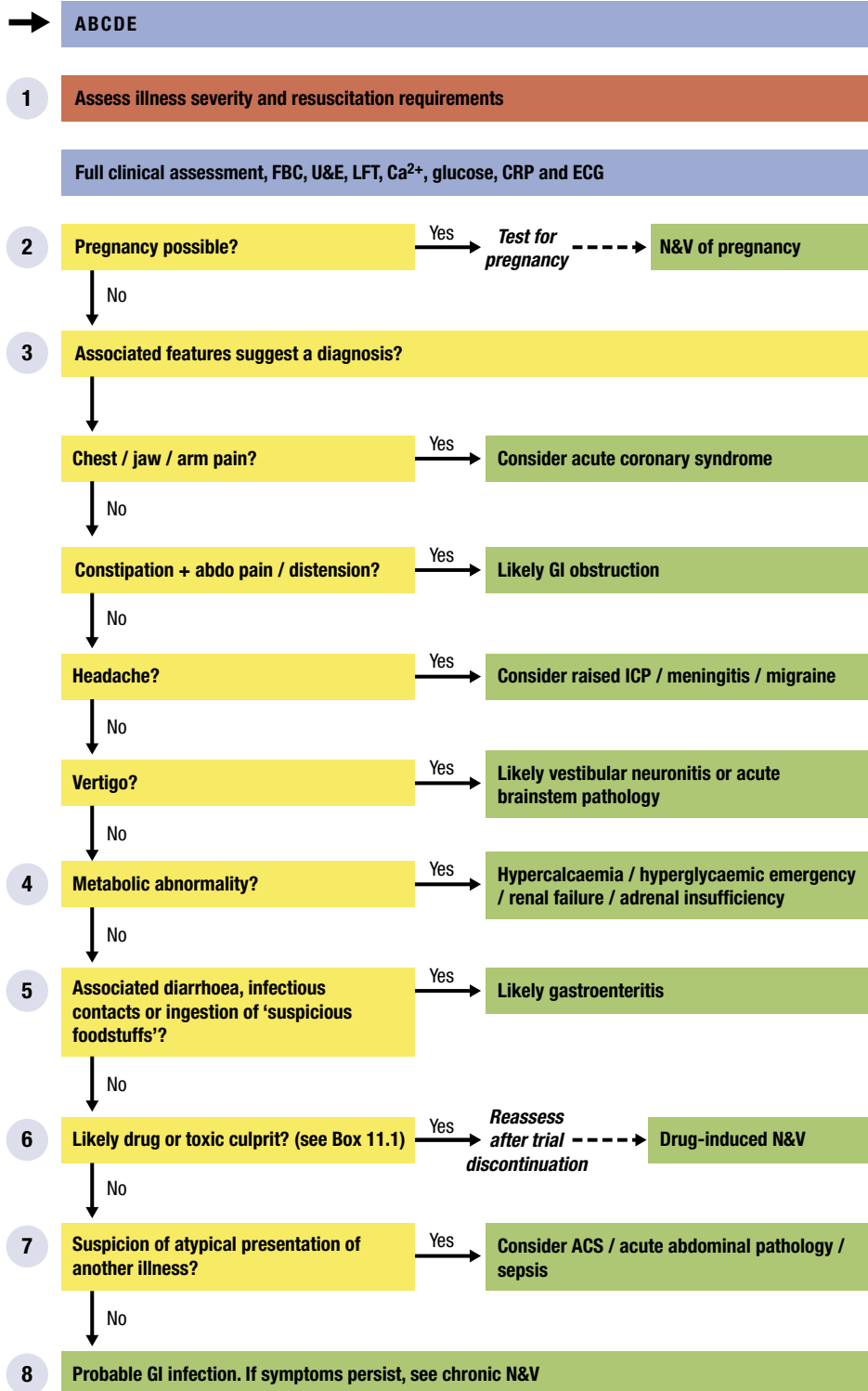
Other causes

Always consider pregnancy in women of child-bearing age with new-onset nausea $\pm$ vomiting. Symptoms are not always confined to morning and may occur at any time of the day.

In bulimia nervosa, recurrent vomiting may be concealed by the patient. There is frequently a history of psychiatric disorder, deliberate self-harm, laxative misuse or evidence of altered body image.

Infections other than gastroenteritis (e.g., urinary tract infection, meningitis, hepatitis or otitis media) may present with vomiting in association with other symptoms, particularly in more vulnerable groups such as children and the elderly.

Myocardial infarction may present with N&V as the predominant symptoms. Historically, this was associated more frequently with inferior infarcts, but recent work suggests it is more likely related to infarct size.



1

Assess illness severity and resuscitation requirements**Step 1 Are there features of shock?**

Look for ↑HR, ↓BP (a late sign) or evidence of tissue hypoperfusion ([Box 2.1](#)). If present, reassess following IV fluid resuscitation and monitor urine output (consider urinary catheterization if necessary). Exclude diabetic ketoacidosis if ↑capillary blood glucose. Suspect adrenal insufficiency in patients with a history of Addison disease (or other autoimmune disease), chronic corticosteroid use or suggestive clinical features ([Box 11.2](#)); also consider whenever the severity of shock appears disproportionate to fluid losses: take blood for random cortisol levels and treat immediately with IV hydrocortisone.

Step 2 Is there acute renal impairment?

Hypovolaemia may result in severe 'prerenal' acute kidney injury (AKI), especially if compounded by antihypertensive or nephrotoxic medication, such as diuretics, ACE inhibitors, NSAIDs. Patients with AKI due to volume depletion require IV rehydration with close monitoring of fluid balance, urine output and electrolytes, particularly potassium. Consider other potential drivers of both AKI and N&V (e.g., sepsis, drug toxicity) in patients without clinical evidence of hypovolaemia. Consider renal failure as the cause of N&V in patients with severe impairment. See [Chapter 29](#) for further assessment of acute kidney injury.

Step 3 Does the patient otherwise require IV fluids/hospital admission?

Patients with clinical evidence of dehydration (thirst, dry mucous membranes, ↓skin turgor) and ongoing vomiting, require IV hydration.

Other features that may indicate a need for hospital admission include:

- Evidence of sepsis ([Chapter 26](#))
- Acute kidney injury ([Chapter 29](#))
- Frail, elderly or immunocompromised patient
- Significant comorbidity, such as heart/renal/hepatic failure

2

Pregnancy possible?

Ascertain gynaecological history and test for pregnancy in any women of reproductive age presenting with N&V.

3

Associated features suggest a diagnosis?

Perform an AXR if N&V is accompanied by absent bowel movements and abdominal pain or distension. Consider gastritis, peptic ulcer disease or acute pancreatitis if there is severe upper abdominal pain and assess as per [Chapter 7](#).

Headache may occur secondary to dehydration in acute vomiting, but consider the possibility of serious CNS pathology. Assess as per [Chapter 13](#) if headache is a prominent feature, especially if the onset is acute.

Acute onset of N&V accompanied by vertigo (dizziness with an illusion of movement, e.g., spinning) is likely to represent either vestibular neuronitis or acute brainstem pathology, such as infarction or haemorrhage; assess as per [Chapter 17](#).

Suspect ACS and assess as per [Chapter 3](#) if N&V occurs with chest/jaw/neck/arm pain and ECG changes.

11

4 Metabolic abnormality?

Test for diabetes and exclude ketoacidosis in any patient without a preexisting diagnosis who has ↑capillary blood glucose (see [Table 25.2](#)). Consider HHS in older patients with markedly ↑ glucose, dehydration, biochemical derangement (↑urea, ↑osmolality) and confusion.

Assess patients with acute kidney injury as described in [Chapter 29](#).

Hypercalcaemia more commonly presents with chronic N&V and is discussed below but may present for the first time with apparent acute N&V. Reassess symptoms after rehydration and other corrective treatment. Investigate the cause of the hypercalcaemia as it may reflect serious underlying disease, such as malignancy.

Consider adrenal insufficiency in patients with ↓Na⁺/↑K⁺ or ↓blood glucose, especially when there are suggestive clinical features ([Box 11.2](#)). If suspected, check a morning cortisol level ± ACTH stimulation test ('short synacthen test').

5 Associated diarrhoea, infectious contacts or ingestion of 'suspicious foodstuffs'?

Suspect acute gastroenteritis if any of these features are present. Most cases are self-limiting viral or toxin-mediated infections and do not require further investigation or antimicrobial treatment. Investigate appropriately if there are concerning features, such as bloody diarrhoea ([Chapter 9](#)), abdominal pain ([Chapter 7](#)) or an excessive systemic inflammatory response ([Box 7.1](#)). Warn patients that they may require further investigation if symptoms persist >10 days (see assessment of chronic N&V below).

6 Likely drug or toxic culprit? ([Box 11.1](#))

Always ask about recent changes to medications or dosages—including over-the-counter medicines. Enquire about alcohol and recreational drug use, particularly cannabis and opiates. In drugs with a narrow therapeutic window (e.g., digoxin, lithium, some anticonvulsants), check serum levels to exclude toxicity. Resolution of symptoms with drug discontinuation (where possible) confirms the diagnosis.

Acute nausea and vomiting: step-by-step assessment

7

Suspicion of atypical presentation of another illness?

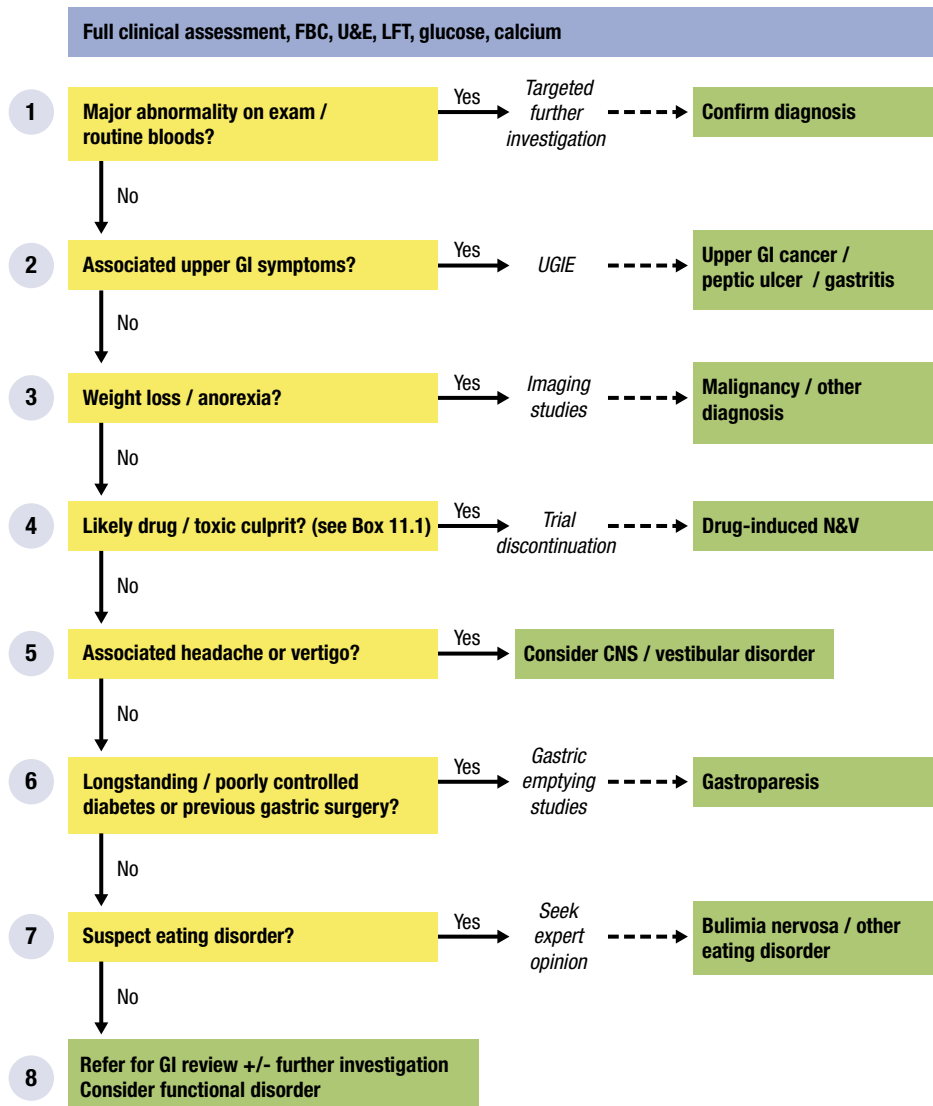
Consider nonspecific presentation of major pathology in patients who appear seriously unwell (e.g., autonomic upset, physiological derangement). Be mindful of atypical presentations in frail elderly patients or those with cognitive impairment, immunocompromise (including chronic corticosteroid treatment) or longstanding diabetes. Perform an ECG to look for evidence of ACS. Undertake a septic screen (see [Chapter 26](#)) if there is fever or ↑inflammatory markers. Have a low threshold for erect CXR, AXR ±

further abdominal imaging, such as CT or USS. Severe constipation may lead to N&V in immobile, frail or elderly patients, so enquire about bowel movements and consider PR exam ± AXR.

8

Probable GI infection

In the absence of alternative pathology, GI infection is the most likely cause of acute N&V; it is almost always self-limiting, so further investigation is usually not required. Reevaluate if the patient is unwell or deteriorating. Proceed to assessment of chronic N&V symptoms persist for >10 days.



Chronic nausea and vomiting: overview

1 Major abnormality on examination/routine blood tests?

Screen patients with persistent or recurrent N&V for the following abnormalities to narrow the differential diagnosis.

Deranged LFTs ± jaundice

Arrange an urgent abdominal USS, review alcohol intake/all recent medications and

assess further, as described in [Chapter 12](#). Continue to look for alternative causes of N&V if only minor LFT derangement without jaundice.

Renal failure

Severe renal failure frequently presents non-specifically with nausea ± vomiting, lethargy and anorexia. Distinguish this from 'prerenal'

AKI due to protracted vomiting and assess for an underlying aetiology ([Chapter 29](#)).

Palpable abdominal or pelvic mass

Arrange urgent imaging to exclude underlying malignancy. Consider USS initially, particularly for young patients. Arrange CT abdomen/pelvis if no cause is found (or to further characterize/stage abnormalities).

Anaemia

Arrange urgent upper GI endoscopy in any patient with persistent N&V and evidence of iron-deficiency anaemia to rule out gastric malignancy. In patients without iron deficiency, continue anaemia work-up ([Chapter 25](#)) but consider other causes of N&V if no clear explanation is identified.

Hypercalcaemia

The critical steps in the management of hypercalcaemia are correcting the calcium and addressing the underlying cause. Check a parathyroid hormone (PTH) level to narrow the differential diagnosis. If PTH is elevated in the context of hypercalcaemia, the likely cause is primary hyperparathyroidism. If the PTH is low (suppressed), then investigate for malignancy (e.g., bone metastasis, myeloma, tumour secretion of PTH-related peptide). To treat hypercalcaemia, rehydrate with IV fluid, and consider a bisphosphonate if the patient remains hypercalcaemic despite fluids. Refer to the appropriate specialty to treat the underlying cause. If mildly $\uparrow$ Ca^{2+} , continue to evaluate for other causes of N&V as small calcium elevations are seen in dehydration.

2 Associated upper GI symptoms?

Associated symptoms of dysphagia, early satiety, epigastric discomfort or dyspepsia suggest upper GI tract pathology such as oesophageal/gastric cancer, peptic ulcer disease or gastritis. Ask about NSAID use/alcohol intake and arrange an UGIE if any of these are present or if vomiting persists for >4 weeks (urgently if weight loss, dysphagia or new onset symptoms in patient aged ≥ 50 years). Refer for formal GI review if persistent symptoms with no cause identified on endoscopy.

3 Weight loss/anorexia?

Investigate for underlying malignancy if N&V is accompanied by significant weight loss (e.g., $\geq 5\%$ of body weight in ≤ 12 months) $\pm$ anorexia, especially in patients with relatively recent onset of symptoms. Arrange an UGIE if not already performed and, if normal, request CT chest/abdomen/pelvis. If no cause identified, continue to assess as below. Consider chronic presentation of adrenal insufficiency if there are other suggestive clinical features ([Box 11.2](#)).

4 Likely drug/toxic culprit?

Take a detailed medication history (including over-the-counter medications, [Box 11.1](#)). Establish alcohol intake and ask specifically about cannabis and/or opiate use. Suspect a toxic aetiology if there is a clear temporal relationship between starting the drug and symptom onset. Reassess after trial discontinuation (if safe to do so) and look for alternative causes if symptoms persist. Check serum levels to exclude toxicity in drugs with a narrow therapeutic window, such as digoxin, lithium, theophylline.

5 Associated headache or neurological symptoms?

Consider $\uparrow$ intracranial pressure if vomiting is associated with recurrent headache that is most prominent on waking from sleep; aggravated by bending, straining or lying; or accompanied by papilloedema—arrange urgent CT of the brain ([Chapter 13](#)).

See [Chapter 17](#) for assessment of patients with concomitant vertigo.

6 Longstanding/poorly controlled diabetes or previous gastric surgery?

Suspect gastroparesis in diabetic patients with longstanding poor control and/or significant microvascular complications or in patients with previous gastric surgery (including weight reduction surgery). Consider prokinetic (i.e. metoclopramide) if no evidence of obstruction/contraindication and refer to GI. Nuclear scintigraphy scan confirms the diagnosis.

7 Suspect eating disorder?

Consider an eating disorder in patients with persistent/recurrent vomiting despite reassuring GI investigations and in the presence of low mood/self-esteem concerning features of body dysmorphia or where family members raise concern. Typically, vomiting occurs after eating and is not directly witnessed. Explore the issue sensitively and, if suspected, seek expert input at an early stage.

8 Refer for GI review/further investigation

Arrange an UGIE (if not already performed) and refer for GI evaluation if symptoms persist without a clear cause. Investigate for chronic pancreatitis if there is associated epigastric discomfort, a history of chronic alcohol excess and/or steatorrhoea; check faecal elastase and arrange abdominal CT. Consider functional vomiting if investigations are consistently reassuring, but be vigilant for underlying eating disorder.

Jaundice 12

Jaundice describes the yellow pigmentation of skin, sclerae and mucous membranes that results from accumulation of bilirubin (hyperbilirubinaemia). It is often classified according to the anatomical location of the pathology:

- Prehepatic—haemolysis
- Hepatic—hepatocellular dysfunction and/or intrahepatic bile duct obstruction
- Posthepatic—extrahepatic bile duct obstruction

It is important to remember that patients may have more than one aetiology (e.g., alcoholic hepatitis on a background of cirrhosis due to chronic hepatitis C infection).

Prehepatic

Haemolytic disorders cause accumulation of unconjugated bilirubin in plasma due to ↑ red cell destruction. LFTs are otherwise normal, but there may be anaemia with evidence of haemolysis on blood tests and film. Clinical features of prehepatic jaundice include normal coloured urine and dark stools. Biochemically there is unconjugated hyperbilirubinaemia. See Clinical Tool: Further assessment of anaemia ([Chapter 25](#)) for further information on the causes and investigation of haemolysis.

Hepatic

Problems with conjugation

Gilbert syndrome is a benign, congenital condition affecting 2–5% of the population. Decreased glucuronyl transferase activity limits bilirubin conjugation and therefore excretion into the bile, causing mild jaundice during periods of fasting or intercurrent illness. There are no other clinical/biochemical features of liver disease.

Acute liver injury

A broad range of liver insults may lead to acute liver injury, such as toxic, infective, autoimmune, metabolic and vascular. Jaundice results from

impaired bilirubin uptake and conjugation by hepatocytes. There may also be obstruction of biliary canaliculi due to inflammation and/or oedema. There is typically a disproportionate increase in ALT and AST relative to ALP and GGT. Extensive liver injury may cause acute liver failure, characterized by encephalopathy ([Table 12.1](#)) and failure of synthetic function (i.e., coagulopathy; typically, INR >1.5) in the absence of preexisting liver disease. Causes of acute liver failure are shown in [Box 12.1](#). Potential causes of acute drug-induced liver injury are listed in [Box 12.2](#). LFTs may take months to normalize after drug cessation, especially with cholestatic injury.

In acute viral hepatitis, jaundice is usually preceded by a 1–2-week prodrome of malaise, arthralgia, headache and anorexia. Hepatitis A–E viruses are responsible for most cases; less common causes include cytomegalovirus (CMV), Epstein–Barr virus (EBV) and herpes simplex. Diagnosis is confirmed by serology. Recovery occurs over 3–6 weeks in most cases, but chronic infection develops in up to 10% of patients with hepatitis B and 80% with hepatitis C.

Autoimmune hepatitis most often presents with established cirrhosis, but 25% of cases manifest as an acute hepatitis with jaundice and constitutional symptoms. It is more common in females (ratio 3:1), and there is an association with other autoimmune conditions. Serum immunoglobulin (IgG) levels are raised, and serum autoantibodies may be present.

Wilson disease, an inherited disorder of copper metabolism, can present with acute hepatitis and occasionally causes acute liver failure. A ↓ serum caeruloplasmin level is highly suggestive.

Hepatic vein thrombosis (Budd–Chiari syndrome) typically presents with upper abdominal pain, hepatomegaly and marked ascites due to liver outflow venous congestion.

Ischaemic hepatitis ('shocked liver') may result from impaired hepatic perfusion in a

Table 12.1 West Haven grading of hepatic encephalopathy

Stage	Alteration of consciousness
0	No change in personality or behaviour No asterixis
1	Impaired concentration and attention span Sleep disturbance, slurred speech Euphoria or depression Asterixis present
2	Lethargy, drowsiness, apathy or aggression Disorientation, inappropriate behaviour, slurred speech
3	Confusion and disorientation, bizarre behaviour Drowsiness or stupor Asterixis usually absent
4	Comatose with no response to voice commands Minimal or absent response to painful stimuli

Box 12.1 Causes of acute liver failure**Drugs/toxins**

- Paracetamol overdose^a
- Anti-tuberculous drugs
- Ecstasy
- Halothane
- *Amanita phalloides*
- Carbon tetrachloride

Infection

- Acute viral hepatitis (A, B, E)^a
- Cytomegalovirus (CMV), Epstein–Barr virus (EBV), herpes simplex

Vascular

- Shocked liver/ischaemic hepatitis
- Budd–Chiari syndrome

Other

- Wilson disease
- Autoimmune hepatitis
- Acute fatty liver of pregnancy
- Extensive malignant infiltration

^aDenotes common cause.

patient with shock; the ALT is usually markedly raised. Cardiac failure may cause hepatic injury due to vascular congestion, with resultant jaundice and ↑ALT.

Cirrhosis

Chronic liver injury results in extensive hepatocellular loss, fibrosis and disturbance of the normal hepatic architecture. Hepatocellular

Box 12.2 Drugs causing acute hepatotoxicity**Acute hepatitis**

- Paracetamol (in overdose)
- Cocaine, ecstasy
- Aspirin, NSAIDs
- Halothane
- Anti-tuberculous therapy: pyrazinamide, isoniazid, rifampicin
- Antifungals: ketoconazole
- Antihypertensives: methyl dopa, hydralazine, dronedarone

Cholestasis/cholestatic hepatitis

- Antibiotics: penicillins, such as flucloxacillin, co-amoxiclav, ciprofloxacin, macrolides, such as erythromycin
- Chlorpromazine
- Azathioprine
- Oestrogens (including the oral contraceptive pill)
- Amitriptyline
- Carbamazepine
- ACE inhibitors
- Cimetidine/ranitidine
- Sulphonamides

Acute alcoholic hepatitis may occur in individuals without chronic liver disease following intensive binge drinking and presents with jaundice, malaise, tender hepatomegaly and fever. The Glasgow Alcoholic Hepatitis Score can be used to predict mortality and guide pharmacological treatment (Table 12.2).

Table 12.2 Glasgow Alcoholic Hepatitis Score (GAHS)

	Score given		
	1	2	3
Age (years)	<50	≥50	—
WCC (10 ⁹ /L)	<15	≥15	—
Urea (mmol/L)	<5	≥5	—
PT ratio or INR	<1.5	1.5–2.0	>2.0
Bilirubin (μmol/L)	<125	125–250	>250

A score of 9 or more identify patients most at risk of death. A score of 9 or more can be used either on day 1 (admission day) or day 6–9.

insufficiency leads to jaundice, coagulopathy and ↓albumin. Architecture disturbance causes portal hypertension; consequent portosystemic shunting of blood results in oesophageal varices and hepatic encephalopathy (Table 12.1). Portal hypertension, ↓albumin and generalized salt and water retention, due to haemodynamic and endocrine abnormalities, lead to ascites. There may be characteristic ‘stigmata’ on examination

Box 12.3 Stigmata of chronic liver disease

- Spider naevi
- Digital clubbing
- Palmar erythema
- Loss of axillary/pubis hair
- Parotid swelling
- Gynaecomastia
- Testicular atrophy

(Box 12.3). Causes of chronic liver injury are listed in Box 12.4.

Hepatic tumours

Malignant infiltration by primary or, more commonly, metastatic tumours may cause jaundice due to intrahepatic duct obstruction or extensive replacement of liver parenchyma. Common associated features include cachexia, malaise, hepatomegaly and RUQ pain (secondary to stretching of the liver capsule).

Post-hepatic/biliary causes of jaundice

Jaundice due to biliary obstruction is associated with a pronounced rise in ALP and GGT (produced in the biliary epithelium). Clinical features may include pale stools, dark urine and intractable itch. A raised prothrombin time (PT) may occur due to vitamin K malabsorption from the GI tract.

Gallstones

Obstructing gallstones are the most common cause of extrahepatic cholestasis; the onset of jaundice is relatively rapid, may be intermittent and is typically accompanied by epigastric/RUQ pain. Cholecystitis is diagnosed on abdominal USS by the appearance of an inflamed gallbladder (thick-walled) containing gallstones. A dilated common bile duct on USS implies extrahepatic biliary obstruction and may be due to gallstones lodged in the common bile duct (choledocholithiasis). The presence of RUQ pain, fever and rigors suggests bacterial infection proximal to the obstruction (ascending cholangitis) and requires prompt antibiotics and decompression (ERCP). More information about gallstones is detailed in Abdominal Pain (Chapter 7).

Benign strictures

Benign strictures may result from trauma, particularly during biliary surgery or as a

Box 12.4 Causes of cirrhosis

- Chronic alcohol excess
- Chronic viral hepatitis (hepatitis B or C)
- Nonalcoholic fatty liver disease
- Autoimmune hepatitis
- Cholestatic
 - Primary sclerosing cholangitis
 - Primary biliary cholangitis
 - Secondary biliary cholangitis
- Metabolic
 - Hereditary haemochromatosis
 - Wilson disease
 - Alpha1-antitrypsin deficiency
 - Cystic fibrosis
- Venous obstruction
 - Cardiac failure
 - Budd–Chiari syndrome
- Drugs, such as methotrexate
- Cryptogenic

consequence of inflammation within the biliary tree, such as recurrent cholangitis, pancreatitis.

Autoimmune

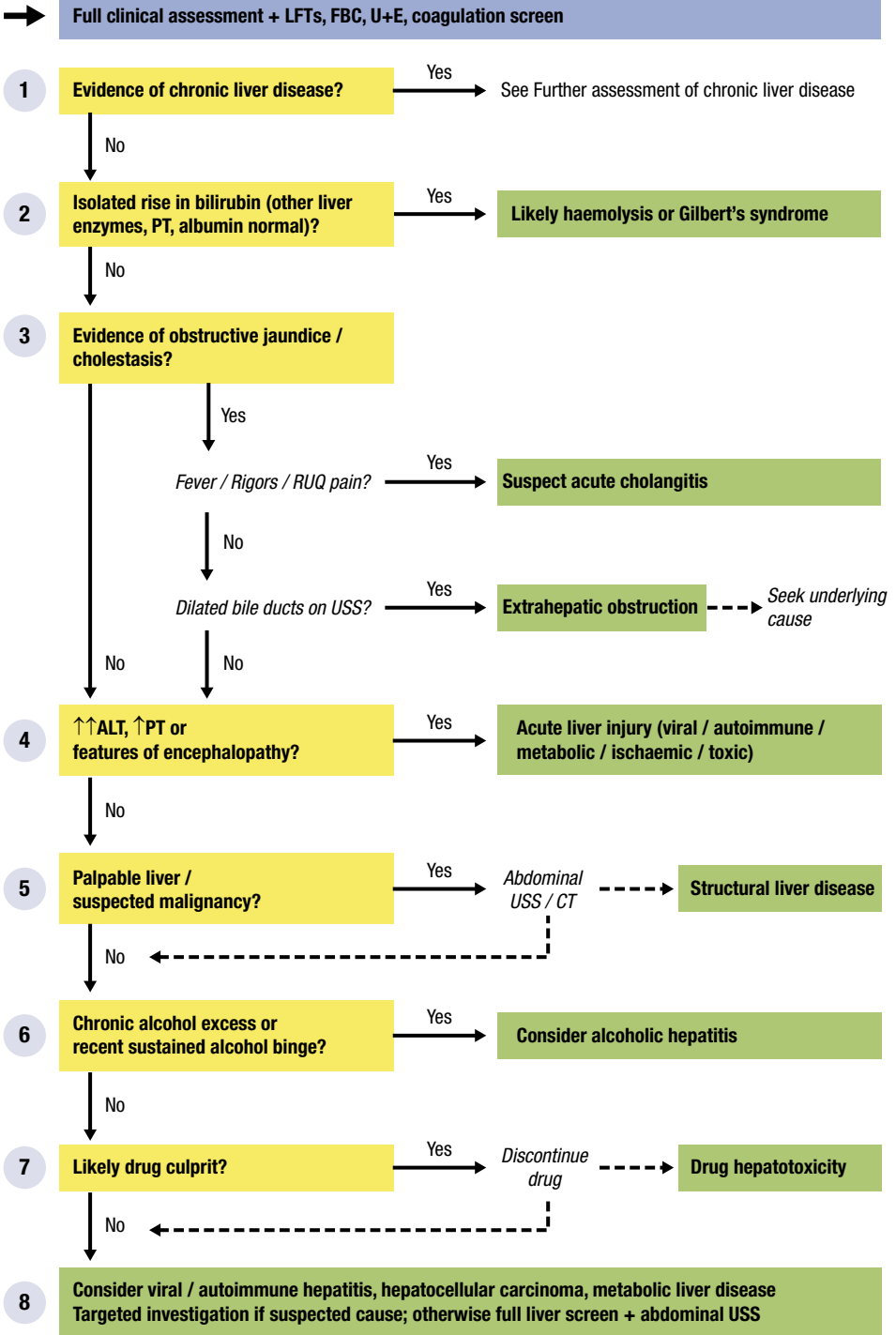
In primary biliary cholangitis (previously referred to as primary biliary cirrhosis), there is progressive destruction of the intrahepatic bile ducts. Ninety percent of patients are female. Pruritus usually precedes jaundice. The presence of antimitochondrial antibodies is diagnostic. Primary sclerosing cholangitis causes inflammation, fibrosis and strictures of the intrahepatic and extrahepatic biliary tree. It is more common in men and has a strong association with ulcerative colitis.

Malignancy

Painless jaundice may represent a malignant cause of obstructive jaundice. Cancer of the head of the pancreas typically presents with insidious, progressive jaundice due to extrinsic compression of the common bile duct, often with marked weight loss, nausea and anorexia. Pain may be absent in the early stages.

Cholangiocarcinoma is a malignant tumour of the intrahepatic or extrahepatic biliary tree that most commonly presents with painless jaundice, often with pruritus.

Other important causes of malignant extrinsic biliary compression include duodenal ampullary tumours and enlarged lymph nodes at the porta hepatis.



1 Evidence of chronic liver disease?

In the absence of an established diagnosis, look for signs of chronic liver disease, particularly physical stigmata (Box 12.3) and evidence of complications of cirrhosis, such as portal hypertension.

- Multiple spider naevi, in the absence of pregnancy/pro-oestrogenic drugs, strongly suggest chronic liver disease. The other stigmata lack specificity individually but are helpful in combination.
- Ascites is a useful pointer to portal hypertension but may have other causes in jaundiced patients, e.g., intra-abdominal malignancy or right heart failure with hepatic congestion. The rapid onset of ascites, jaundice, hepatomegaly and abdominal pain suggests acute thrombotic complication, that is, Budd–Chiari syndrome.
- Hepatic encephalopathy (Table 12.1) and synthetic dysfunction (see below) may occur in acute liver failure but are highly suggestive of chronic liver disease when they occur against a background of previous clinical/biochemical evidence of liver disturbance.
- Abdominal USS can demonstrate morphologic features of cirrhosis (e.g., coarse echotexture, ↑nodularity) and provide supportive evidence of portal hypertension (splenomegaly, reversed flow in the portal vein).

Evaluate any patient with jaundice and a background of established or suspected cirrhosis as described in Further assessment of chronic liver disease below.

2 Isolated rise in bilirubin (other liver enzymes, PT, albumin normal)?

Consider haemolysis if features of liver disease are absent and blood results suggest ↑red cell breakdown (e.g., ↓Hb, ↑LDH, ↓haptoglobin, red cell fragments on blood film) ± evidence of ↑red cell production (e.g., ↑reticulocytes, polychromasia).

If there are no features of haemolysis or chronic liver disease, jaundice is mild (bilirubin

<100 µmol/L) and LFTs are otherwise normal, the likely diagnosis is Gilbert syndrome.

3 Evidence of obstructive jaundice/cholestasis?

A rise in ALP/GGT out of proportion to ALT/AST suggests that jaundice is due to impaired biliary excretion. If there are features of sepsis, take blood cultures, give IV antibiotics, arrange urgent abdominal USS and discuss with the surgical team. Otherwise, perform USS to look for bile duct dilatation (indicating extrahepatic obstruction) and an underlying cause, such as gallstones.

If USS does not show dilated bile ducts, proceed to step 4.

If USS shows dilated bile ducts but does not reveal the underlying cause, arrange further imaging. MRCP is the gold standard modality for biliary tree imaging—request this if you suspect gallstones (acute or recurrent RUQ pain) or if there is a history of biliary pathology. CT allows better visualization of the pancreas—arrange if pancreatic cancer is suspected, e.g., jaundice is painless with an insidious, progressive onset, especially in the elderly.

4 ↑↑ALT, ↑PT or features of encephalopathy?

Identify patients with acute liver failure

Look for features of acute liver failure in all jaundiced patients without preexisting liver disease or extrahepatic biliary obstruction, especially if there is evidence of extensive hepatocellular injury (↑↑ALT).

Hepatic encephalopathy (Table 12.1) may be subtle; look closely for ↓concentration/alertness, mild disorientation, behavioural changes and reversal of the sleep/wake cycle. Test specifically for constructional apraxia (e.g., ask the patient to draw a five-pointed star or clock face) and asterixis. Seek collateral history if possible and exclude hypoglycaemia and other metabolic abnormalities. Consider CT brain to exclude intracranial pathology, especially if there are focal neurological abnormalities.

Step-by-step assessment

An ↑PT (in the absence of anticoagulation, preexisting coagulopathy or extensive cholestasis) indicates ↓hepatic synthetic function. If ↑, do not correct acutely unless there is major haemorrhage.

Seek cause of acute liver injury

Enquire about alcohol and paracetamol intake, all recent drugs (prescribed or otherwise) and possible exposure to environmental/occupational toxins, e.g., carbon tetrachloride. Check viral serology (IgM antibodies to hepatitis B core antigen [HBc], hepatitis A virus [HAV], hepatitis E virus [HEV], CMV and EBV), serum caeruloplasmin, autoantibodies (ANA, ASMA, LKM) and gamma-globulins, and paracetamol level (on admission blood draws). Arrange abdominal USS to exclude Budd–Chiari syndrome and extensive malignant infiltration.

Repeatedly monitor and reassess patients with acute liver failure

Monitor closely in ICU or a high-dependency unit (HDU) for complications, including:

- Hypoglycaemia
- Hyperkalaemia
- Metabolic acidosis
- Renal failure (develops in >50%, often necessitating haemofiltration)
- Cerebral oedema with ↑intracranial pressure
- Bacterial or fungal infection

Use the King's College Hospital criteria (Box 12.5) to identify patients who may require transplantation. Liaise early with a specialist liver unit to enable timely transfer if necessary.

5 Palpable liver/suspected malignancy?

Consider abdominal USS or CT to exclude malignant or other structural liver disease if the patient has:

- Clinical evidence of hepatomegaly
- Known malignancy with metastatic potential
- A palpable abdominal/rectal mass or lymphadenopathy
- Ascites in the absence of chronic liver disease
- Insidious constitutional upset, such as unexplained weight loss/anorexia/cachexia.

Box 12.5 King's College Hospital criteria for liver transplantation in acute liver failure

Paracetamol overdose

- pH <7.25 at or beyond 24 hours following the overdose
or
- Serum creatinine >300 µmol/L or anuria *plus* PT >100 seconds (INR > 6.5) *plus* encephalopathy grade 3 or 4

Nonparacetamol cases

Severe encephalopathy and

- PT >100 seconds (INR >6.5)
or
- Any three of the following:
 - Jaundice to severe encephalopathy time >7 days
 - Age <10 or >40 years
 - Indeterminate (non-A, non-B hepatitis) or drug-induced causes
 - Bilirubin >300 µmol/L
 - PT >50 seconds (INR >3.5)

Creatinine of 300 µmol/L $\cong$ 3.38 mg/dL. Bilirubin of 300 µmol/L $\cong$ 17.6 mg/dL.

Modified from O'Grady JG, Alexander GJ, Hayllar KM, Williams R. Early indicators of prognosis in fulminant hepatic failure. Gastroenterology. 1989;97(2):439–445.

6 Chronic alcohol excess or recent sustained alcohol binge?

Establish alcohol intake in all jaundiced patients. A tactical approach is to ask a general enquiry, before quantifying typical weekly intake. In addition to the weekly total number of drinks, consider the alcohol content and, for spirits, the measure per drink. Seek a collateral history from relatives and friends and observe closely for features of alcohol withdrawal if you suspect covert alcohol abuse.

Suspect alcoholic hepatitis in any patient with longstanding excessive consumption, e.g., >40 units/week or recent binge drinking (e.g., >100 units/week), or who develops severe withdrawal symptoms after 24–48 hours in hospital. Helpful supporting features for alcoholic hepatitis include tender hepatomegaly and an AST:ALT ratio of >1. Once the diagnosis is made, calculate the Glasgow alcoholic hepatitis score to assess prognosis and guide treatment (Table 12.2).

7 Likely drug culprit?

Enquire about all drugs taken within the preceding 6 weeks, including those which are prescribed, recreational and over-the-counter, such as NSAIDs, paracetamol and herbal remedies. Box 12.2 contains some common causes of hepatotoxicity, but there are many others; consult with a pharmacist or review the literature if uncertain. Wherever possible, discontinue any suspected drug culprit and observe the effect on bilirubin and LFTs. If paracetamol toxicity is suspected, immediate treatment with acetylcysteine may be required; treat as per a poison information resource such as TOXBASE® (www.toxbase.org).

Liver biochemistry may take months to normalize in drug-induced liver injury, so consider further investigations (see below) in the interim unless clinical suspicion of drug toxicity is very high, for example, the patient recently started co-amoxiclav or anti-tuberculous therapy.

8 Consider other causes. Targeted investigation or full liver screen

If the cause remains uncertain, conduct a basic liver screen for viral, autoimmune, hereditary and metabolic conditions (Box 12.6), arrange

an abdominal USS if not already performed and confirm alcohol intake. If there is fever, purpura, ↓platelets, conjunctival congestion or recent exposure to potentially contaminated water (e.g., freshwater sports, sewage worker), send blood and urine samples for leptospiral culture, serology ± PCR (liaise with the Microbiology laboratory) and consider empirical antibiotic treatment. Liver biopsy may be required if jaundice persists without a clear cause and potential drug culprits have been removed.

Box 12.6 Screening investigations in jaundice/suspected liver disease

Serology

- Hepatitis B surface antigen (HBsAg), hepatitis C antibodies HCV Ab (+ IgM antibodies to HBc, HAV, HEV, CMV and EBV if acute)

Metabolic

- Ferritin, alpha-1 antitrypsin level, caeruloplasmin

Autoimmune

- AMA, ASMA, ANA, LKM, immunoglobulins

Other

- Abdominal USS, alpha-fetoprotein, paracetamol level, toxicology screen

Further assessment of chronic liver disease

Step 1 Establish the underlying cause

Determine the underlying aetiology in all new presentations of cirrhosis. Assess alcohol intake and perform screening investigations as listed in [Box 12.6](#). Exclude causes with specific interventions (e.g., Wilson disease, hepatitis virus B and C, haemochromatosis). Consider referral for liver biopsy if the cause remains uncertain.

Step 2 Look for evidence of complications/decompensation

Evaluate for hepatic encephalopathy, portal hypertension and other complications of liver disease; examine for confusion, ascites, oedema, jaundice and malnutrition. Measure albumin and PT to assess hepatic synthetic function, as well as bilirubin, U+Es (hyponatraemia, hepatorenal syndrome) and FBC (anaemia, thrombocytopenia). Consider investigation for pulmonary complications such as pleural effusion, hepatopulmonary syndrome and pulmonary hypertension in patients with breathlessness, cyanosis or $\downarrow\text{SpO}_2$. If not already done, screen for oesophageal varices with UGIE and for hepatocellular carcinoma with 6–12 monthly abdominal USS; USS will also detect small-volume ascites and splenomegaly.

Step 3 Seek precipitants of decompensation

Patients with cirrhosis have limited metabolic reserve, so many factors can precipitate acute decompensation, including spontaneous bacterial peritonitis, other intercurrent infection, surgery, alcohol excess, hepatotoxic drugs, constipation and hepatocellular carcinoma.

In a decompensated cirrhotic patient (i.e., acute worsening of jaundice, coagulopathy or encephalopathy):

- Perform a full sepsis screen including an ascitic tap for microscopy and culture if ascites is present.
- Review all drugs and alcohol intake.
- Ensure that an abdominal USS has been performed within the last 6 months.

Additionally, in patients with encephalopathy:

- Seek evidence of upper GI bleeding (stool evaluation, $\downarrow\text{Hb}$, $\uparrow\text{urea}$).
- Look for and correct constipation (PR examination, stool chart)/dehydration/electrolyte disturbance.
- Minimize use of opioids/other sedative drugs or drugs that may precipitate constipation.

In cases of worsening ascites, evaluate salt and water intake and compliance with diuretics, exclude spontaneous bacterial peritonitis, and consider Budd–Chiari syndrome.

Step 4 Assess prognosis

Use the Child–Pugh classification ([Table 12.3](#)) to assess prognosis in patients with cirrhosis. Other poor prognostic factors include $\uparrow\text{creatinine}$ and $\downarrow\text{Na}^+$, a small liver, and variceal haemorrhage. Consider referral to a transplant centre for patients with Child–Pugh grade B or C. Absolute contraindications to transplantation include severe extrahepatic disease with a predicted mortality of $>50\%$ at 5 years, severe irreversible pulmonary disease, active malignancy (excluding specific neuroendocrine malignancies), and active alcohol/illicit drug abuse. Ongoing extrahepatic sepsis and untreated HIV are also absolute contraindications, but discussion with a Liver Transplant unit would be merited given their possible temporary nature.

Table 12.3 Child–Pugh classification of cirrhosis

Score	1	2	3
Bilirubin ($\mu\text{mol/L}$)	<34	34–51	>51
Albumin (g/L)	>35	28–35	<28
Ascites	None	Mild	Marked
Encephalopathy (Table 12.1)	None	Grade 1–2	Grade 3–4
Prolongation of PT (seconds)	<4	4–6	>6
Child A = Score <7 : 2-year survival 85% Child B = Score 7–9: 2-year survival 60% Child C = Score >9 : 2-year survival 35%			